

Network Layer

Functions of Network Layer

Addressing:

- Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in network.

Packeting:

- This is performed by Internet Protocol. The network layer converts the packets from its upper layer.

Routing:

- It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.

Inter-networking:

- It works to deliver a logical connection across multiple devices.

IP Addresses

IP Addresses: Classful Addressing

What is an IP Address?

***An IP address is a
32-bit
address.***

Note

***The IP addresses
are
unique.***

Address space rule

**The address space in a protocol
That uses N-bits to define an
Address is:**

$$2^N$$

IPv4 address space

The address space of IPv4 is

2^{32}

or

4,294,967,296.

Binary Notation

01110101 10010101 00011101 11101010

Dotted-decimal notation

10000000

00001011

00000011

00011111

128.11.3.31

Example 1

Change the following IP address from binary notation to dotted-decimal notation.

10000001 00001011 00001011 11101111

Solution

129.11.11.239

Example 2

Change the following IP address from dotted-decimal notation to binary notation:

111.56.45.78

Solution

01101111 00111000 00101101 01001110

Example 3

Find the error in the following IP Address
111.56.045.78

Solution

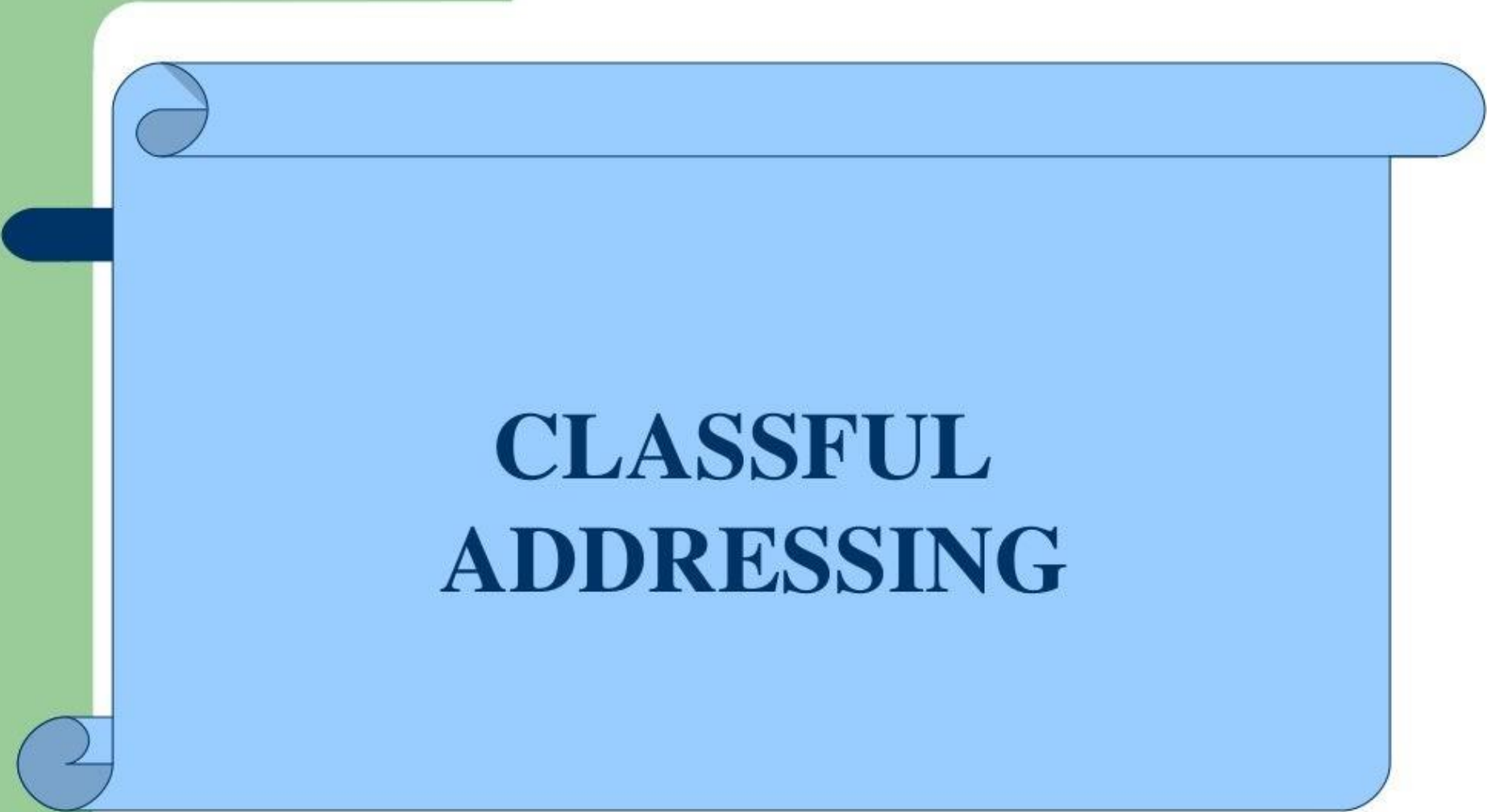
There are no leading zeroes in
Dotted-decimal notation (045)

Example 3 (continued)

Find the error in the following IP Address
75.45.301.14

Solution

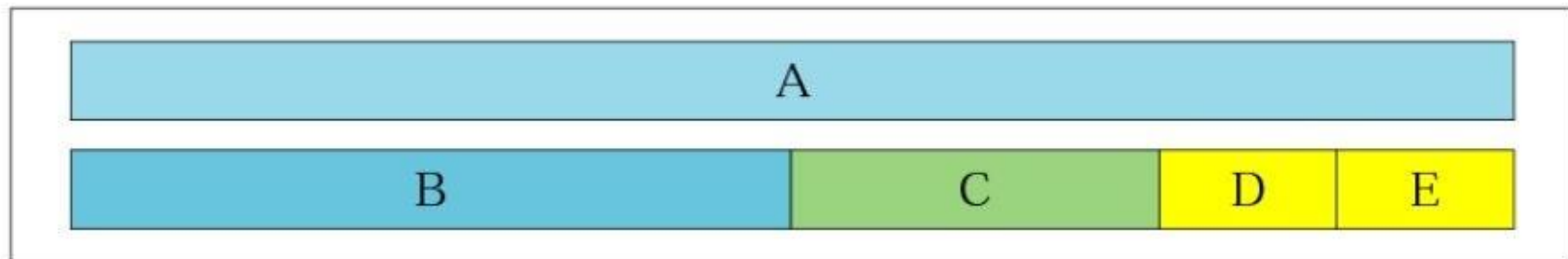
In decimal notation each number ≤ 255
301 is out of the range



CLASSFUL ADDRESSING

Occupation of the address space

Address space



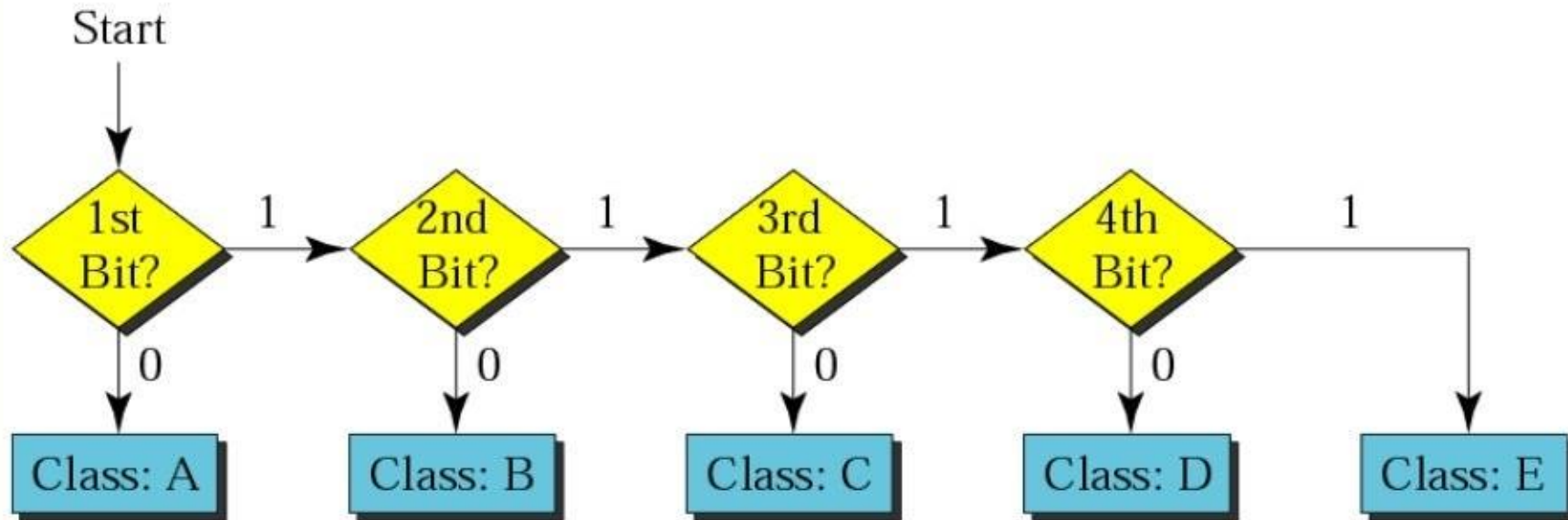
In classful addressing the address space is divided into 5 classes:

A, B, C, D, and E.

Finding the class in binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

Finding the address class



Example 5

Show that Class **A** has
 $2^{31} = 2,147,483,648$ addresses

Example 6

Find the class of the following IP addresses

00000001 00001011 00001011 11101111
11000001 00001011 00001011 11101111

Solution

- 00000001 00001011 00001011 11101111
1st is 0, hence it is Class A
- 11000001 00001011 00001011 11101111
1st and 2nd bits are 1, and 3rd bit is 0 hence, Class C

Finding the class in decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

Example 7

Find the class of the following addresses

158.223.1.108

227.13.14.88

Solution

- 158.223.1.108

1st byte = 158 ($128 < 158 < 191$) class B

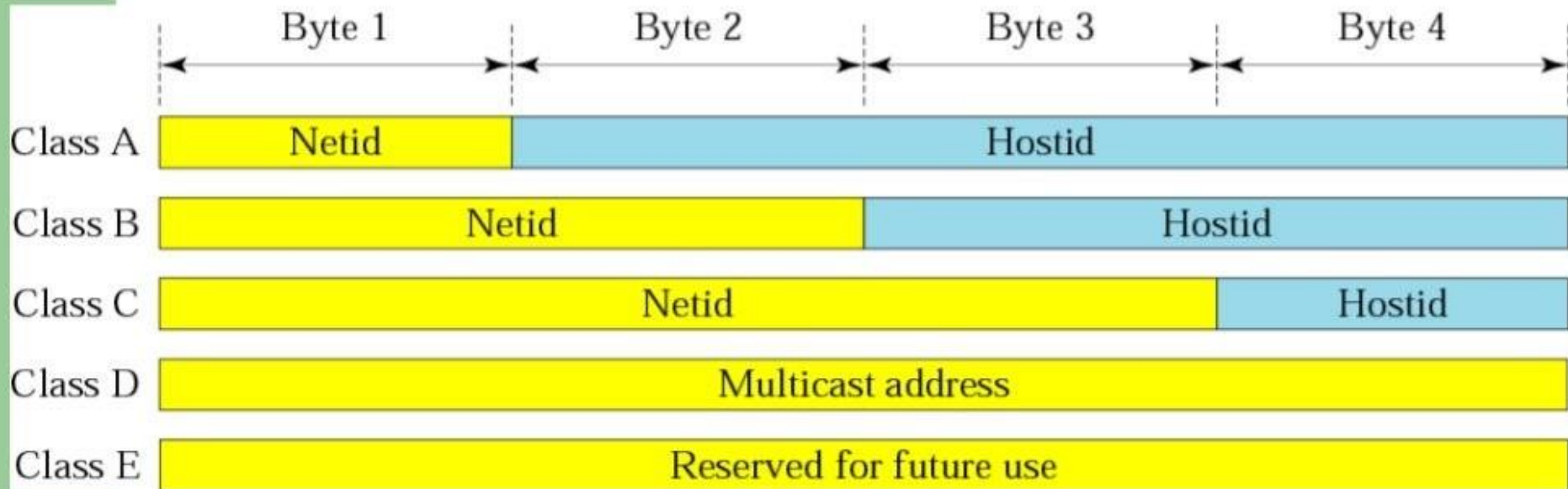
- 227.13.14.88

1st byte = 227 ($224 < 227 < 239$) class D

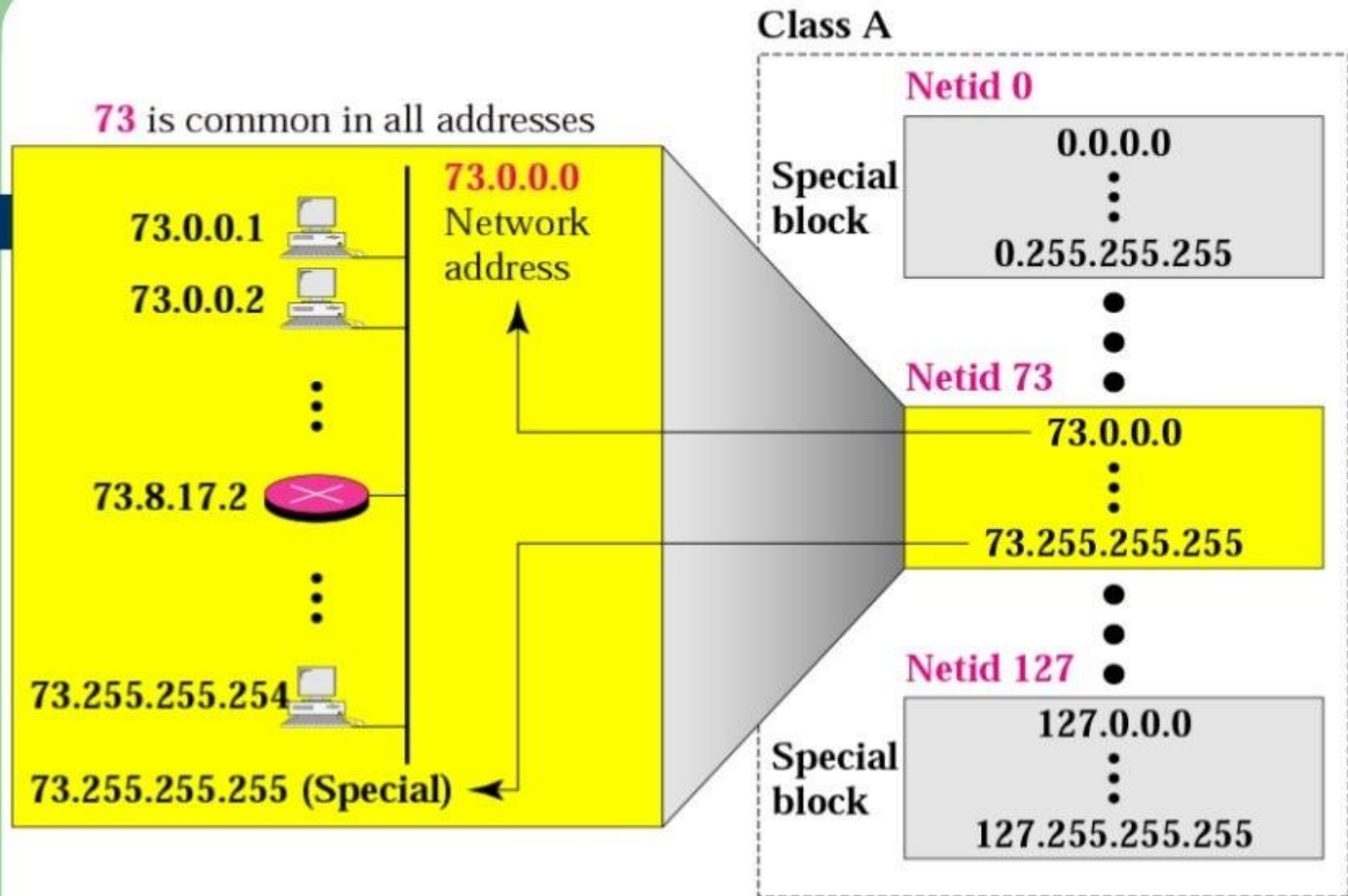
IP address with appending port number

- 158.128.1.108:25
- the four octets before colon is the IP address
- The number of colon (25) is the port number

Netid and hostid



Blocks in class A



128 blocks: 16,777,216 addresses in each block

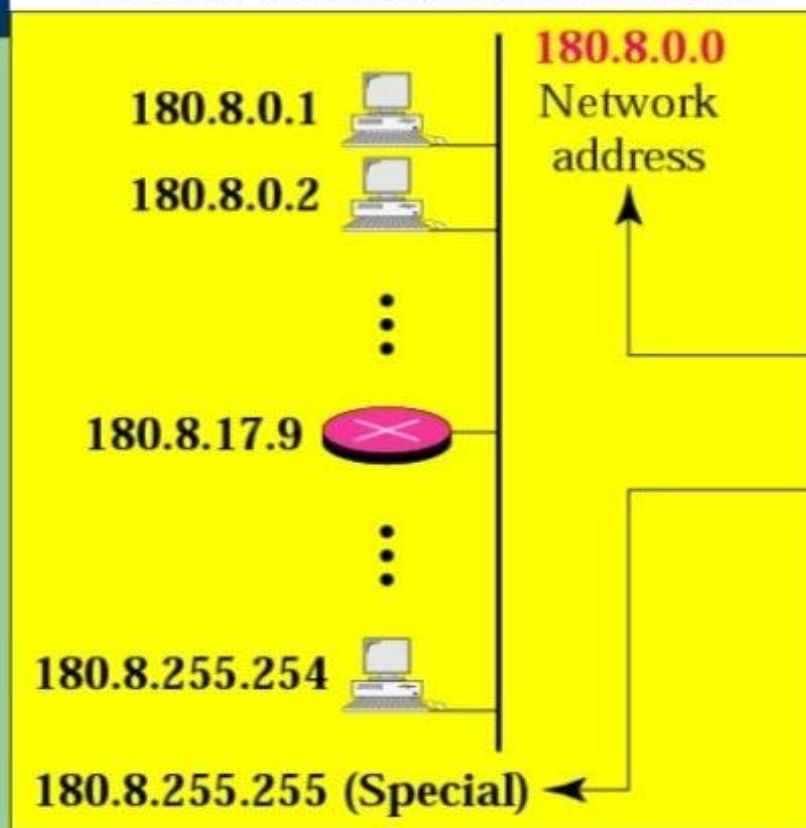


Note

***Millions of class A addresses
are wasted.***

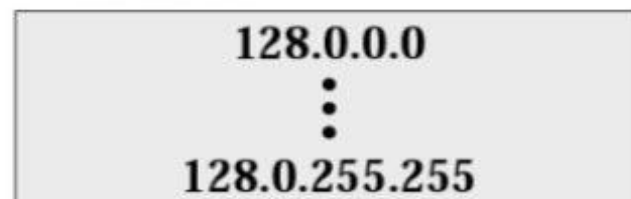
Blocks in class B

180.8 is common in all addresses

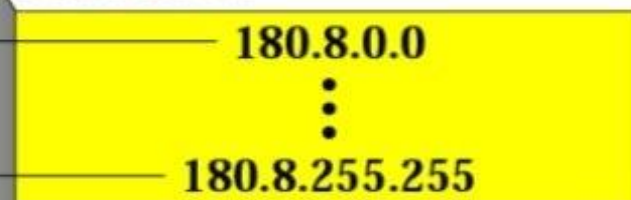


Class B

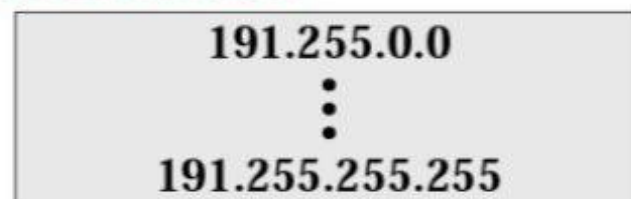
Netid 128.0



Netid 180.8



Netid 191.255

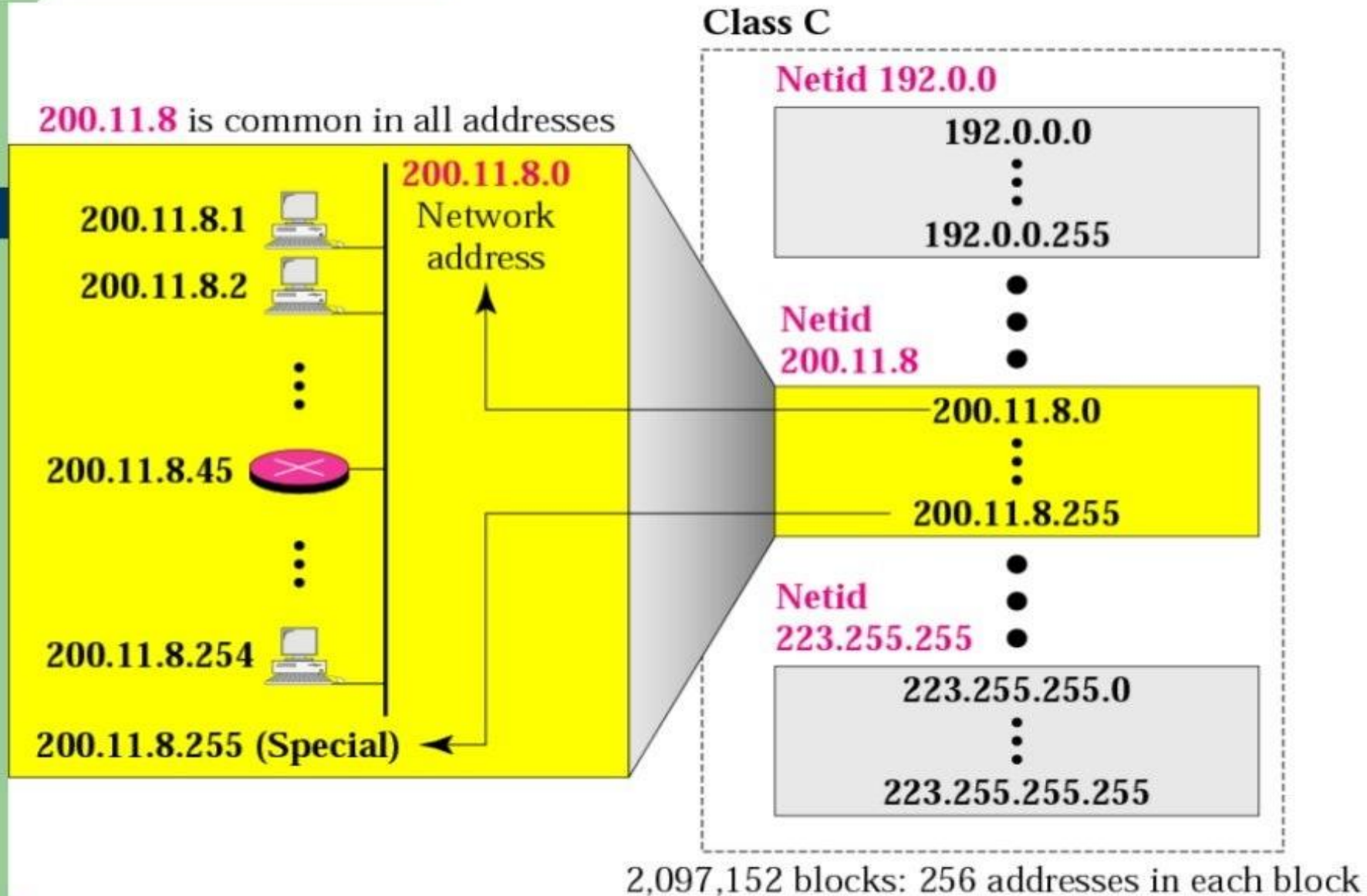


16,384 blocks: 65,536 addresses in each block

Note

***Many class B addresses
are wasted.***

Blocks in class C



Note

***The number of addresses in
a class C block
is smaller than
the needs of most organizations.***

Note

***Class D addresses
are used for multicasting;
there is only
one block in this class.***

Note

Class E addresses are reserved for special purposes; most of the block is wasted.

Network Addresses

The network address is the first address.

The network address defines the network to the rest of the Internet.

Given the network address, we can find the class of the address, the block, and the range of the addresses in the block

Note

***In classful addressing,
the network address
(the first address in the block)
is the one that is assigned
to the organization.***

Example 8

Given the network address 132.21.0.0, find the class, the block, and the range of the addresses

Solution

The 1st byte is between 128 and 191.

Hence, Class B

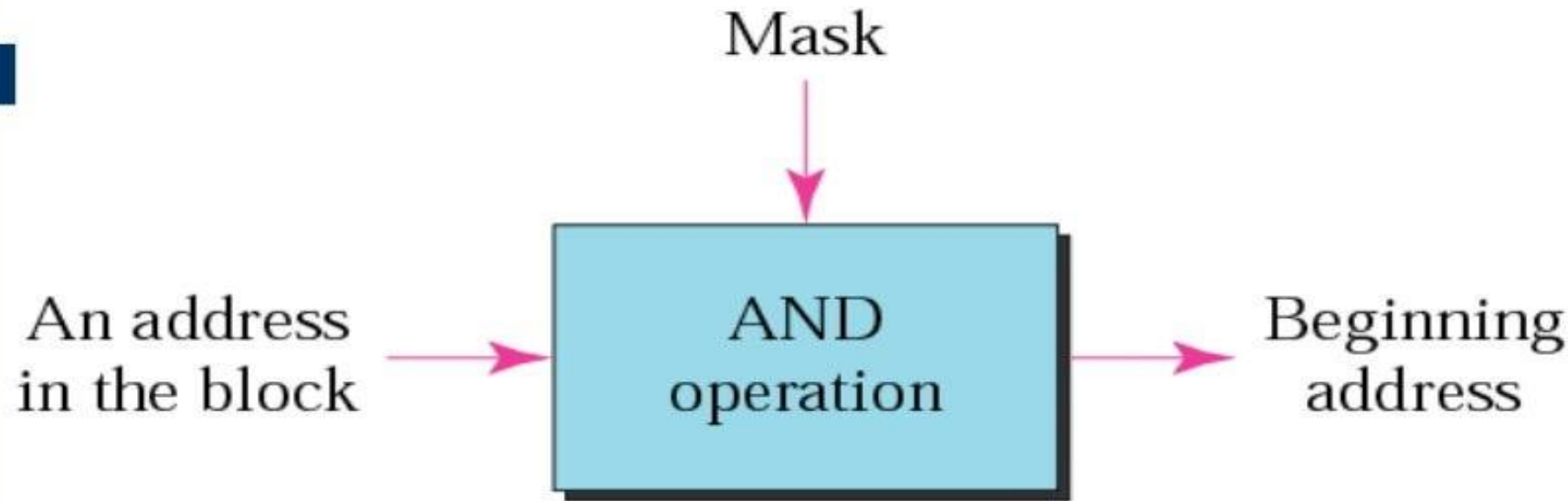
The block has a netid of 132.21.

The addresses range from
132.21.0.0 to 132.21.255.255.

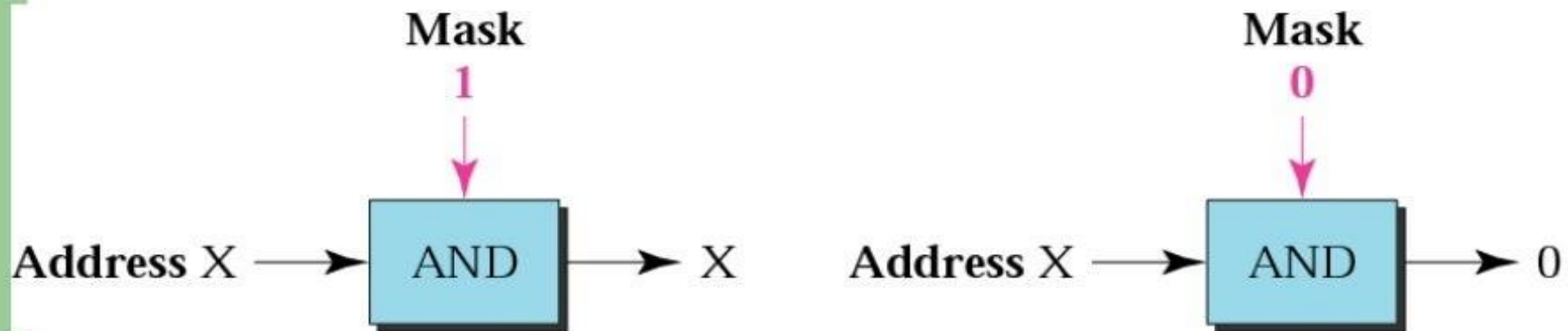
Mask

- A mask is a 32-bit binary number.
- The mask is **ANDed** with IP address to get
 - The block address (Network address)
 - **Mask And IP address = Block Address**

Masking concept



AND operation



Note

The network address is the beginning address of each block. It can be found by applying the default mask to any of the addresses in the block (including itself). It retains the **netid** of the block and sets the **hostid** to zero.

Default Mak

- Class A default mask is 255.0.0.0
- Class B default mask is 255.255.0.0
- Class C Default mask 255.255.255.0

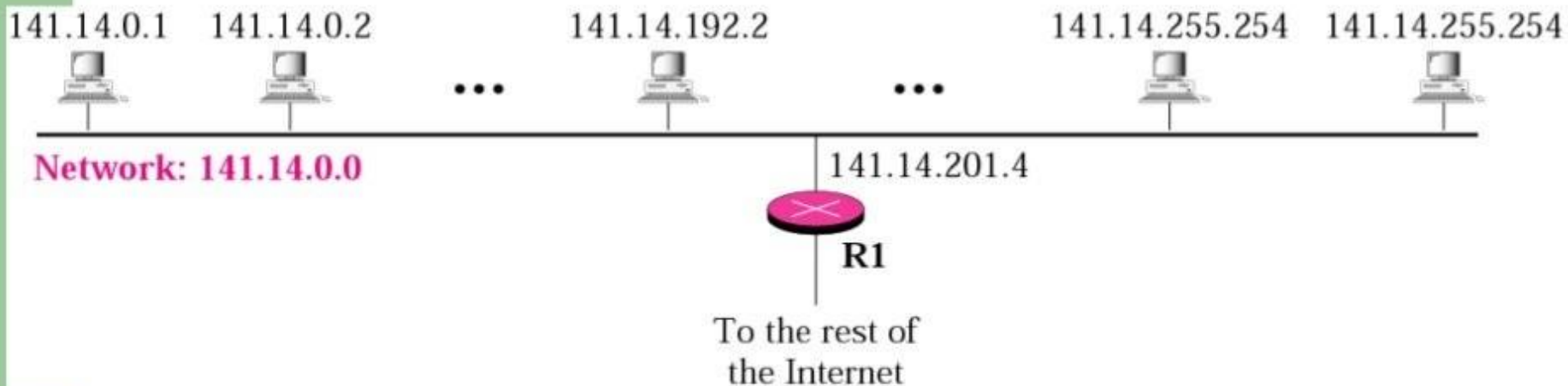
5.1

SUBNETTING

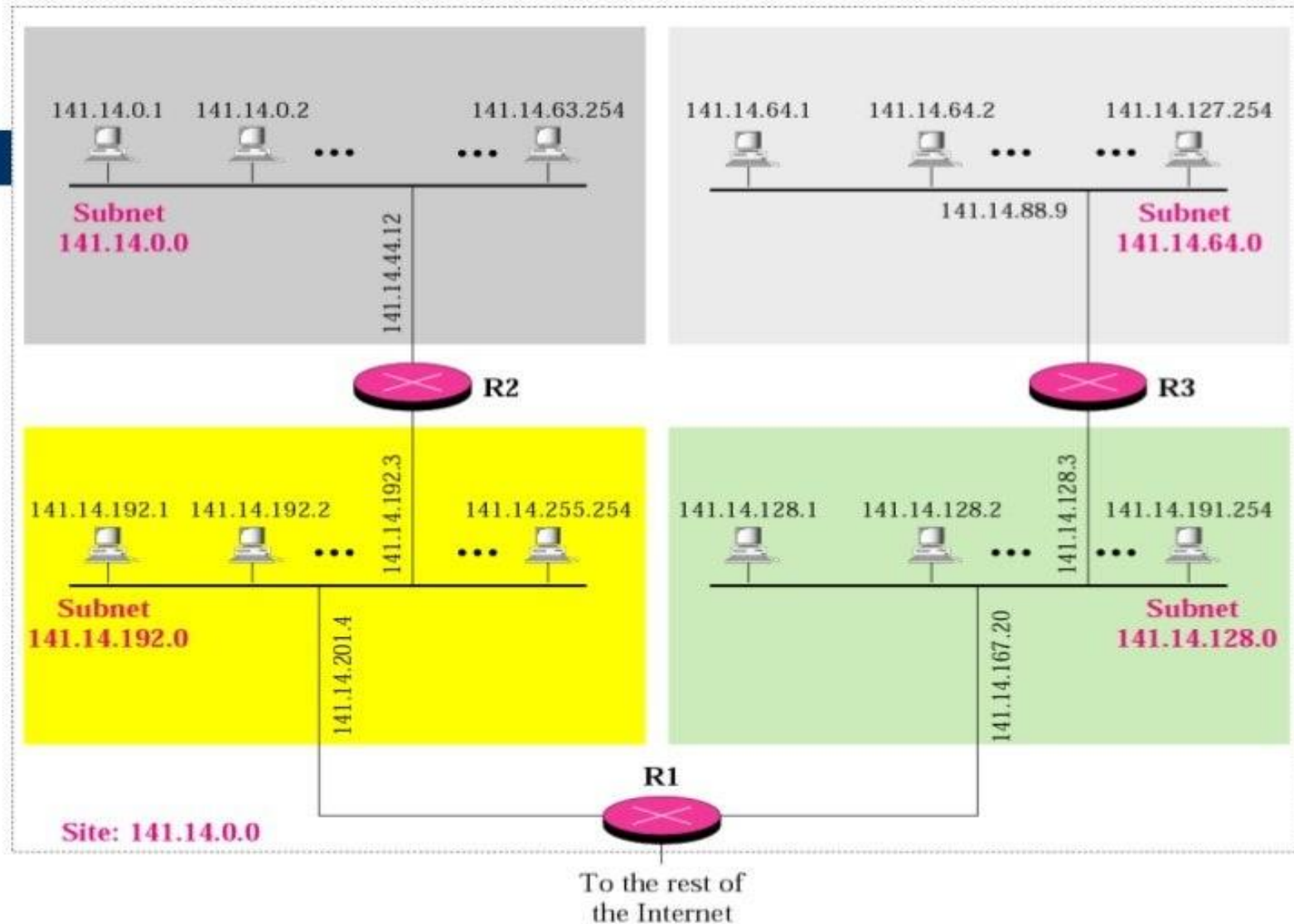
Note

IP addresses are designed with two levels of hierarchy.

A network with two levels of hierarchy (not subnetted)



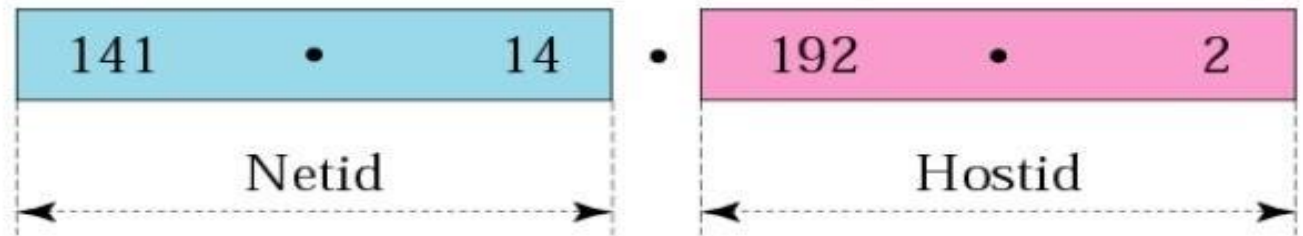
A network with three levels of hierarchy (subnetted)



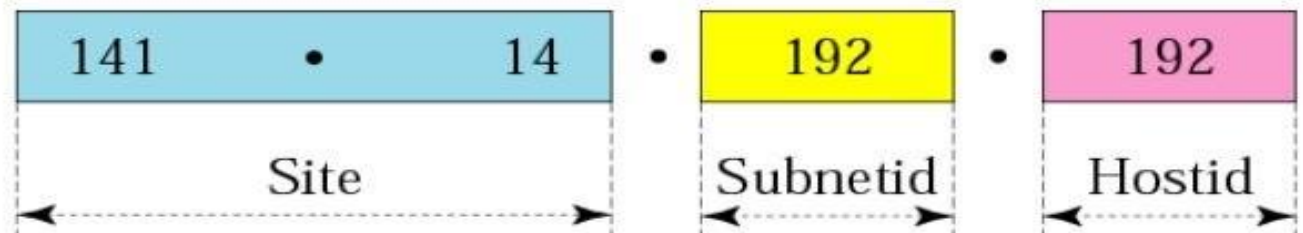
Note

- Subnetting is done by borrowing bits from the host part and add them the network part

Addresses in a network with and without subnetting

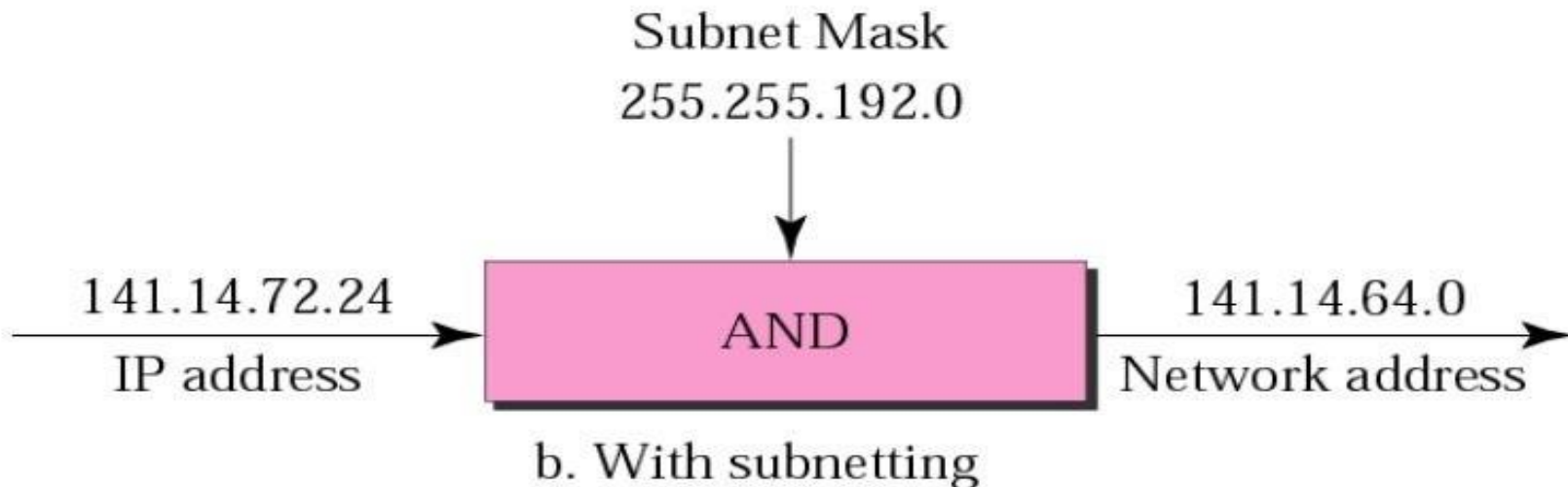
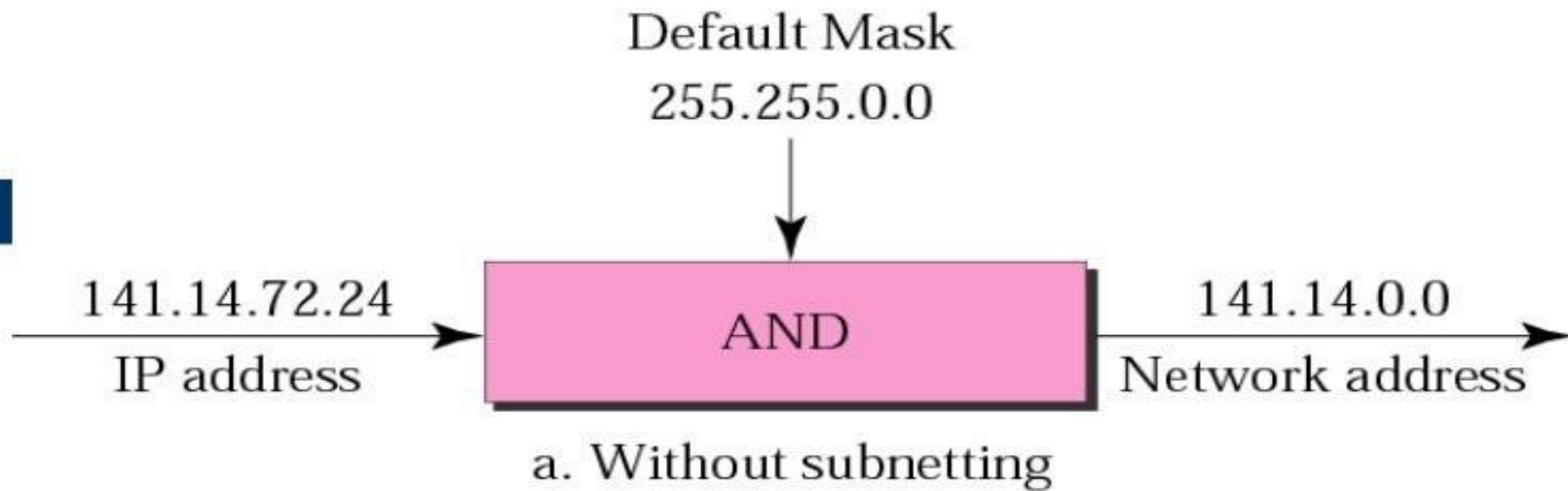


a. Without subnetting



b. With subnetting

mask and subnet mask



Finding the Subnet Address

Given an IP address, we can find the subnet address the same way we found the network address. We apply the mask to the address. We can do this in two ways: straight or short-cut.

Straight Method

In the straight method, we use binary notation for both the address and the mask and then apply the AND operation to find the subnet address.

Example 9

What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?

Solution

11001000 00101101 00100010 00111000

11111111 11111111 11110000 00000000

11001000 00101101 00100000 00000000

The subnetwork address is **200.45.32.0**.

Short-Cut Method

- *** If the byte in the mask is 255, copy the byte in the address.
- *** If the byte in the mask is 0, replace the byte in the address with 0.
- *** If the byte in the mask is neither 255 nor 0, we write the mask and the address in binary and apply the AND operation.

Example 10

What is the subnetwork address if the destination address is 19.30.80.5 and the mask is 255.255.192.0?

Solution

See next slide

Solution

IP Address

19	•	30	•	84	•	5
----	---	----	---	----	---	---

Mask

255	•	255	•	192	•	0
-----	---	-----	---	-----	---	---

19	•	30	•	64	•	0
----	---	----	---	----	---	---

Subnet Address

84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0

Comparison of a default mask and a subnet mask

	255.255.0.0			
Default Mask	11111111 11111111		00000000 00000000	
			16	
	255.255.224.0			
Subnet Mask	11111111 11111111		111	00000 00000000
			3	13

Note

The number of subnets must be a power of 2.

Example 11

A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.

Solution

The number of 1s in the default mask is 24 (class C).

Solution (Continued)

The company needs six subnets. This number 6 is not a power of 2. The next number that is a power of 2 is 8 (2^3). We need 3 more 1s in the subnet mask. The total number of 1s in the subnet mask is 27 ($24 + 3$).

The total number of 0s is 5 ($32 - 27$). The mask is

Solution (Continued)

11111111 11111111 11111111 11100000

or

255.255.255.224

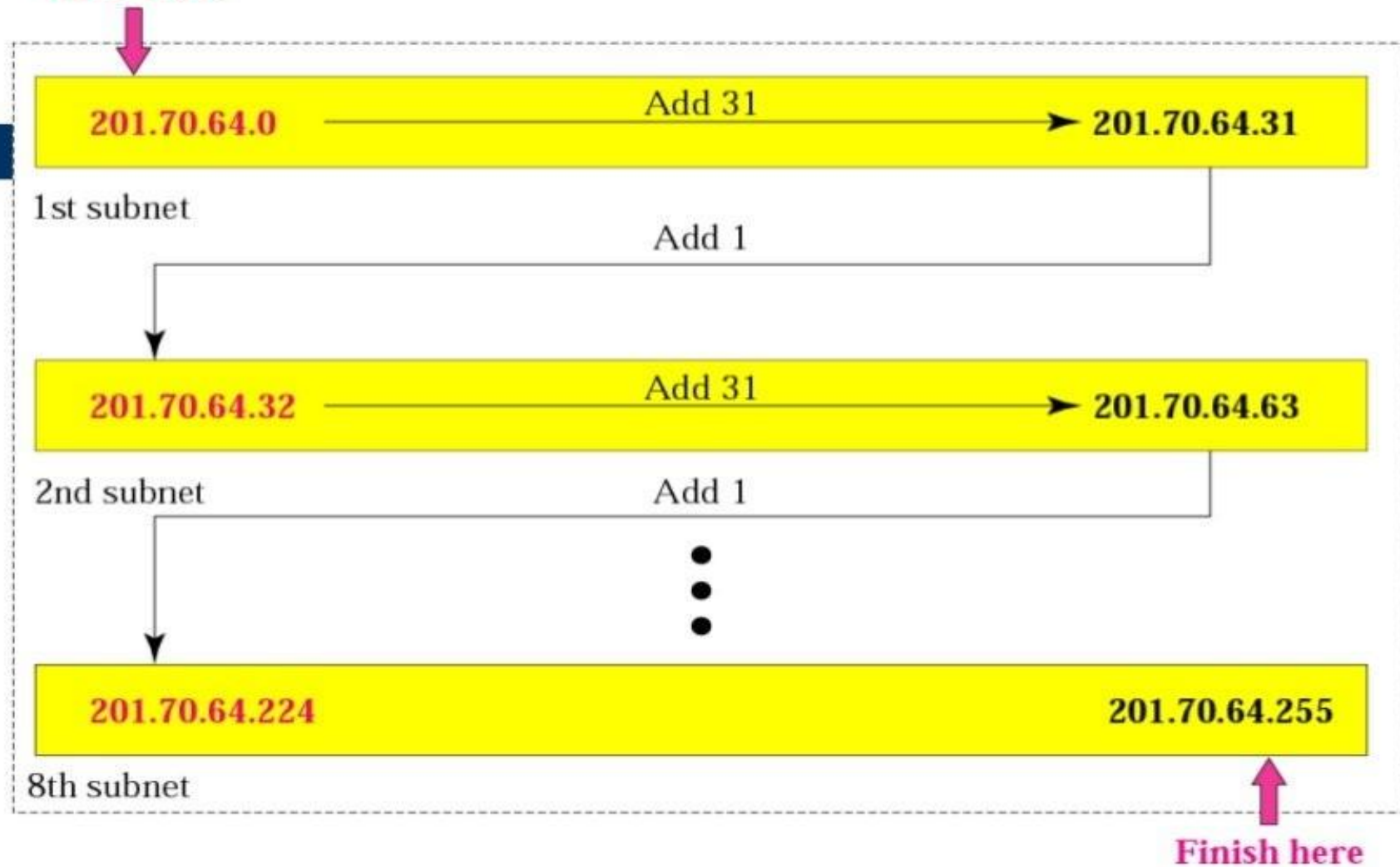
The number of subnets is 8.

The number of addresses in each subnet is 2^5 (5 is the number of 0s) or 32.

See Next slide

Example 3

Start here



Example 12

A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets.

Solution

The number of 1s in the default mask is 16 (class B).

Solution (Continued)

The company needs 1000 subnets. This number is not a power of 2. The next number that is a power of 2 is 1024 (2^{10}). We need 10 more 1s in the subnet mask.

The total number of 1s in the subnet mask is 26 ($16 + 10$).

The total number of 0s is 6 ($32 - 26$).

Solution (Continued)

The mask is

11111111 11111111 11111111 11000000

or

255.255.255.192.

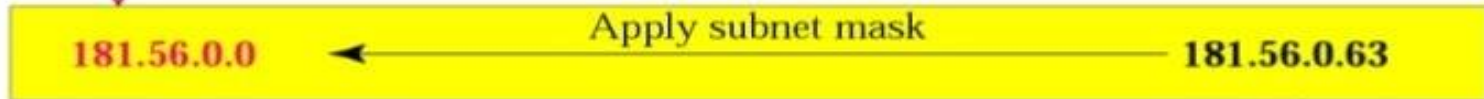
The number of subnets is 1024.

The number of addresses in each subnet is 2^6
(6 is the number of 0s) or 64.

See next slide

Example 4

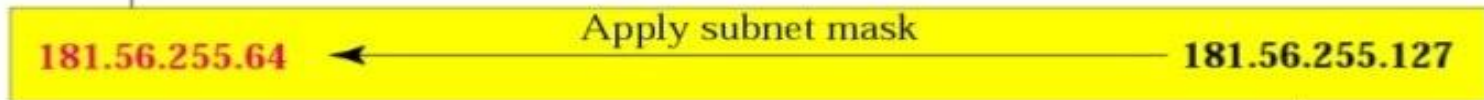
Finish here



1st subnet

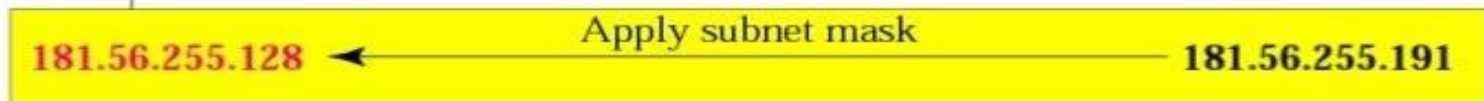
•
•
•

subtract 1



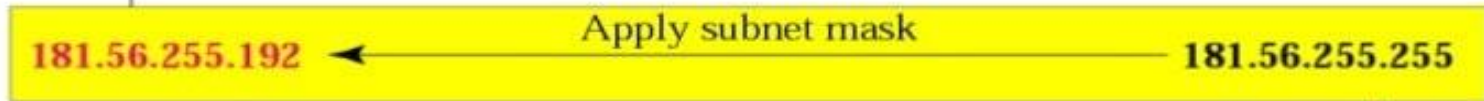
1022th subnet

subtract 1



1023th subnet

subtract 1



1024th subnet



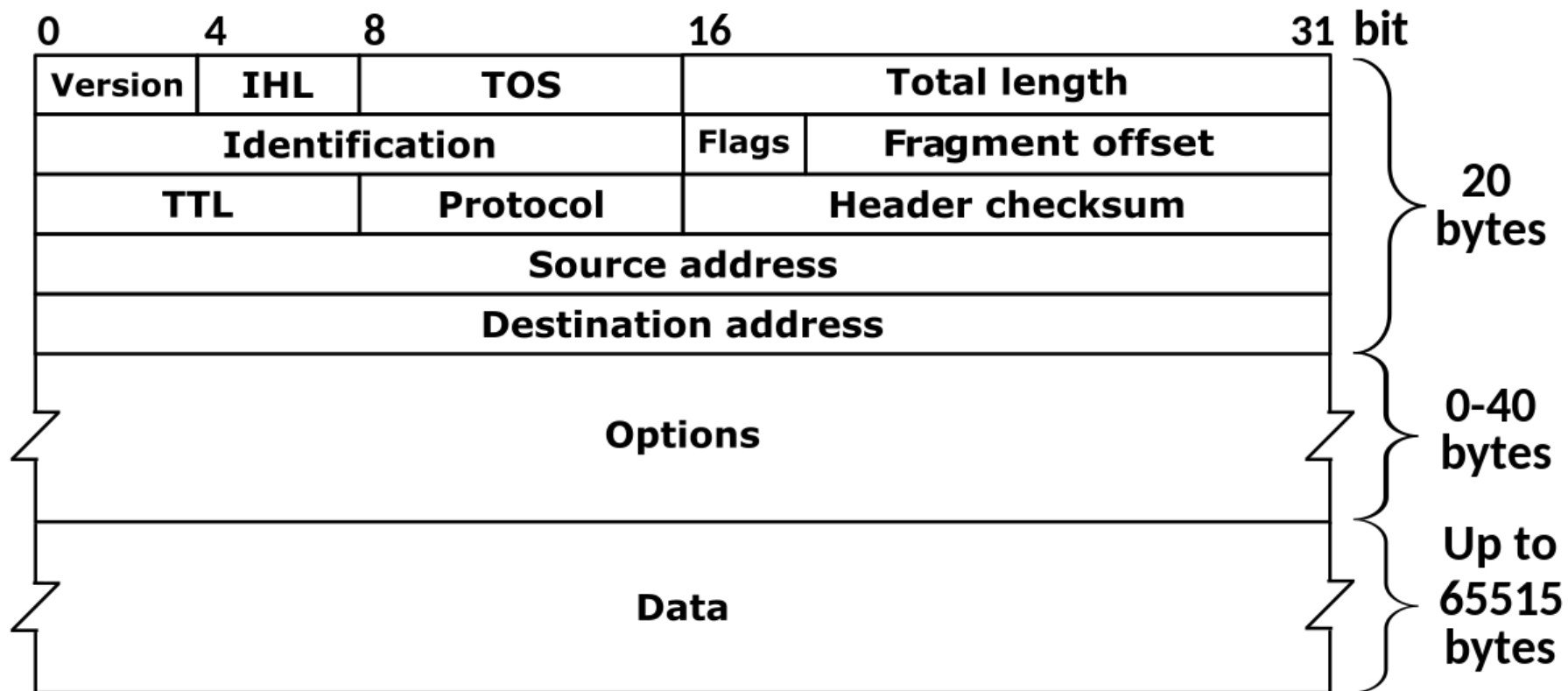
Start here

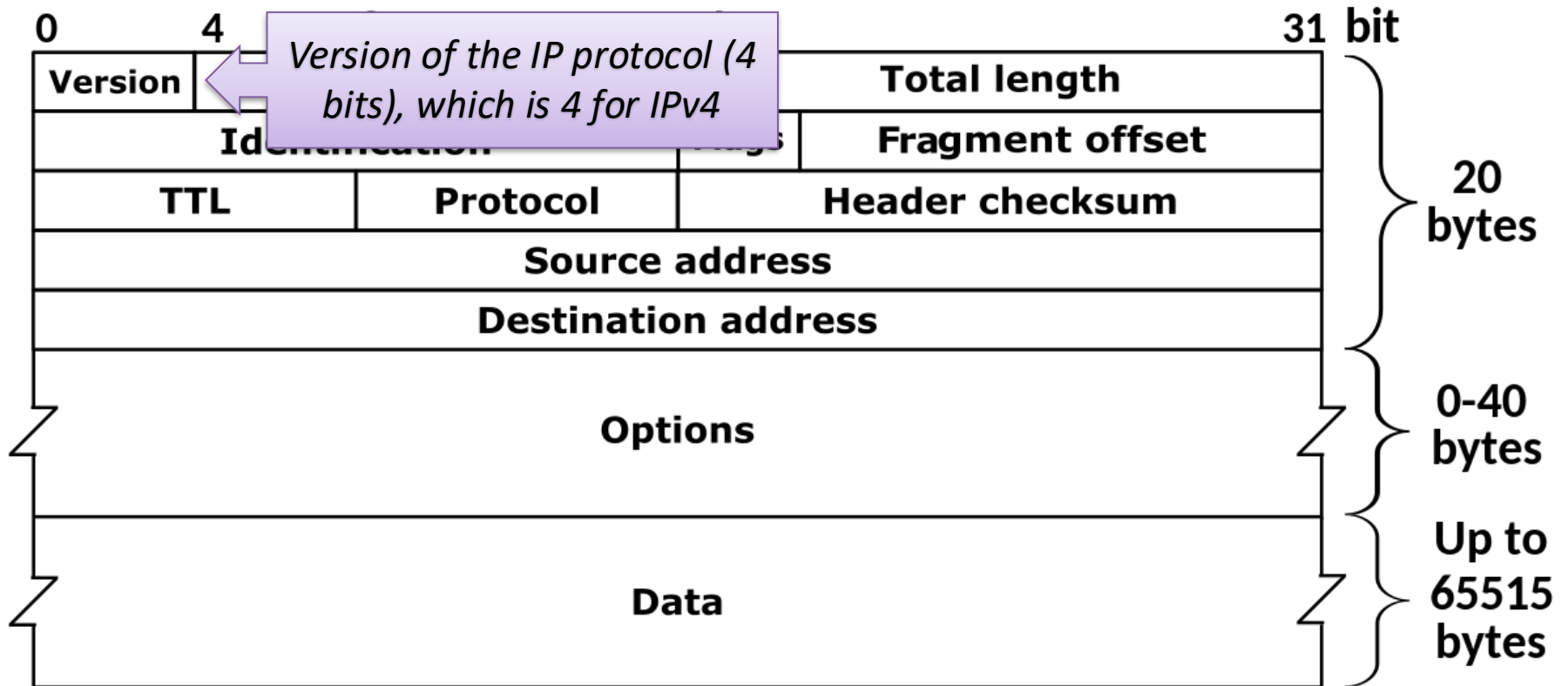
Slash notation

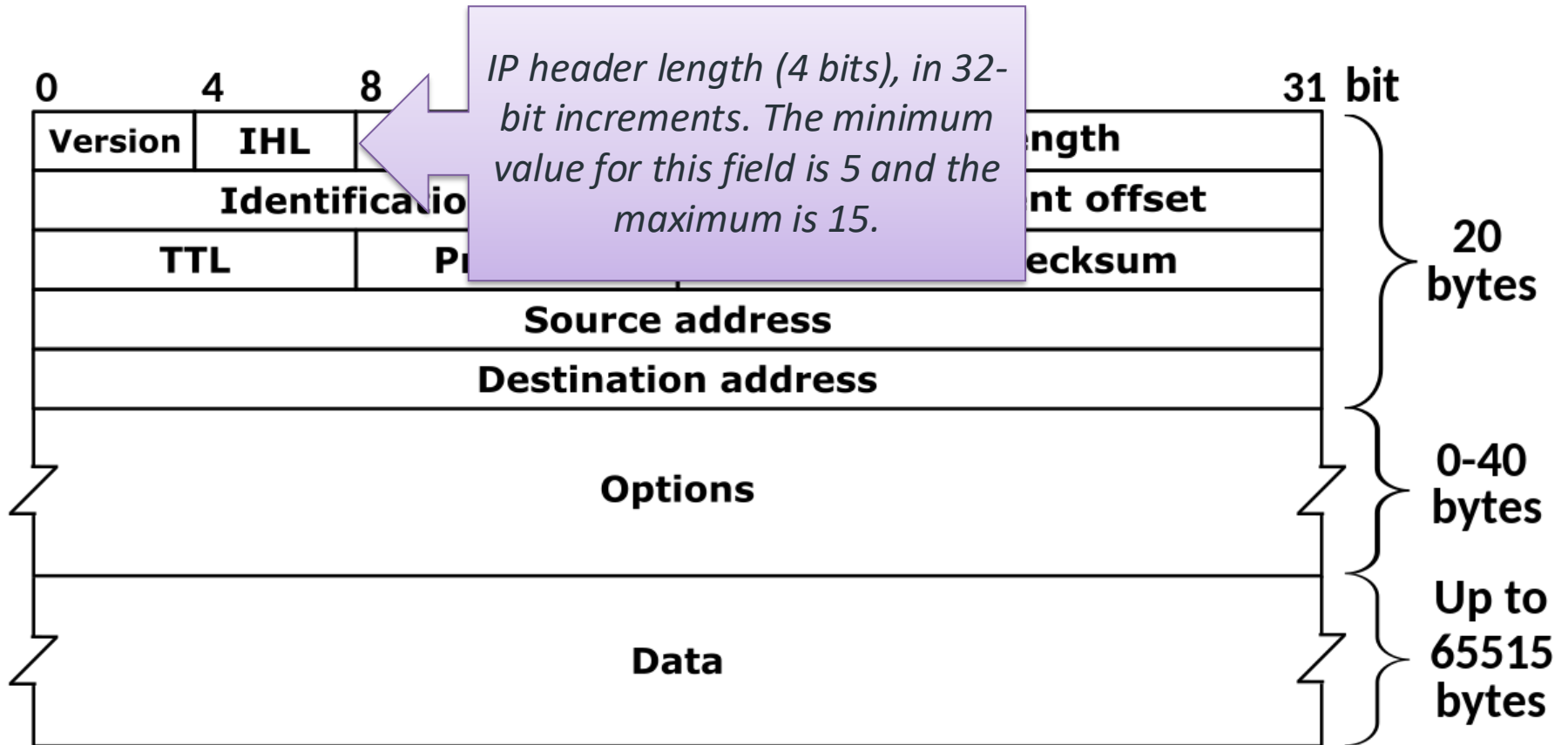
A.B.C.D/*n*

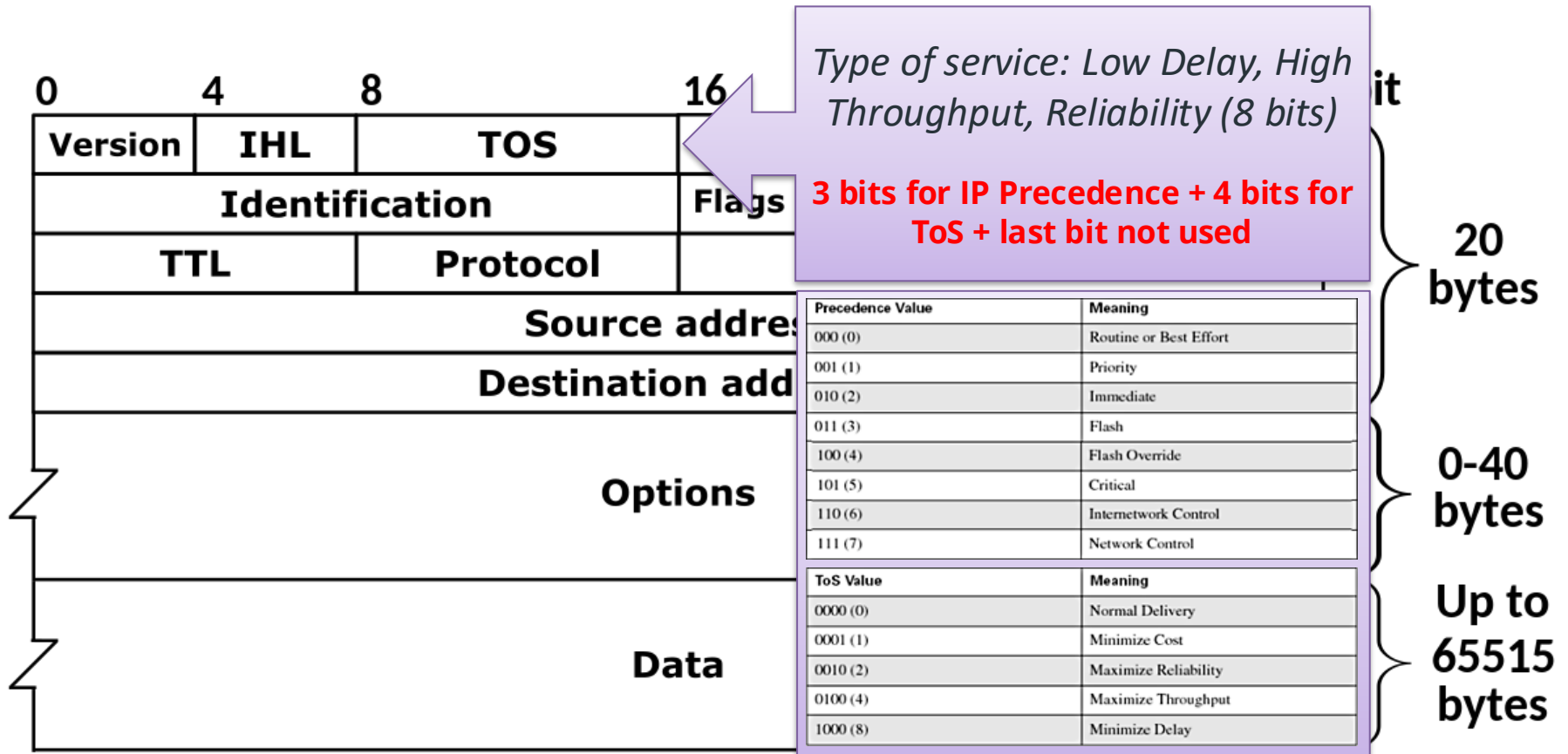
IPv4 Header Format

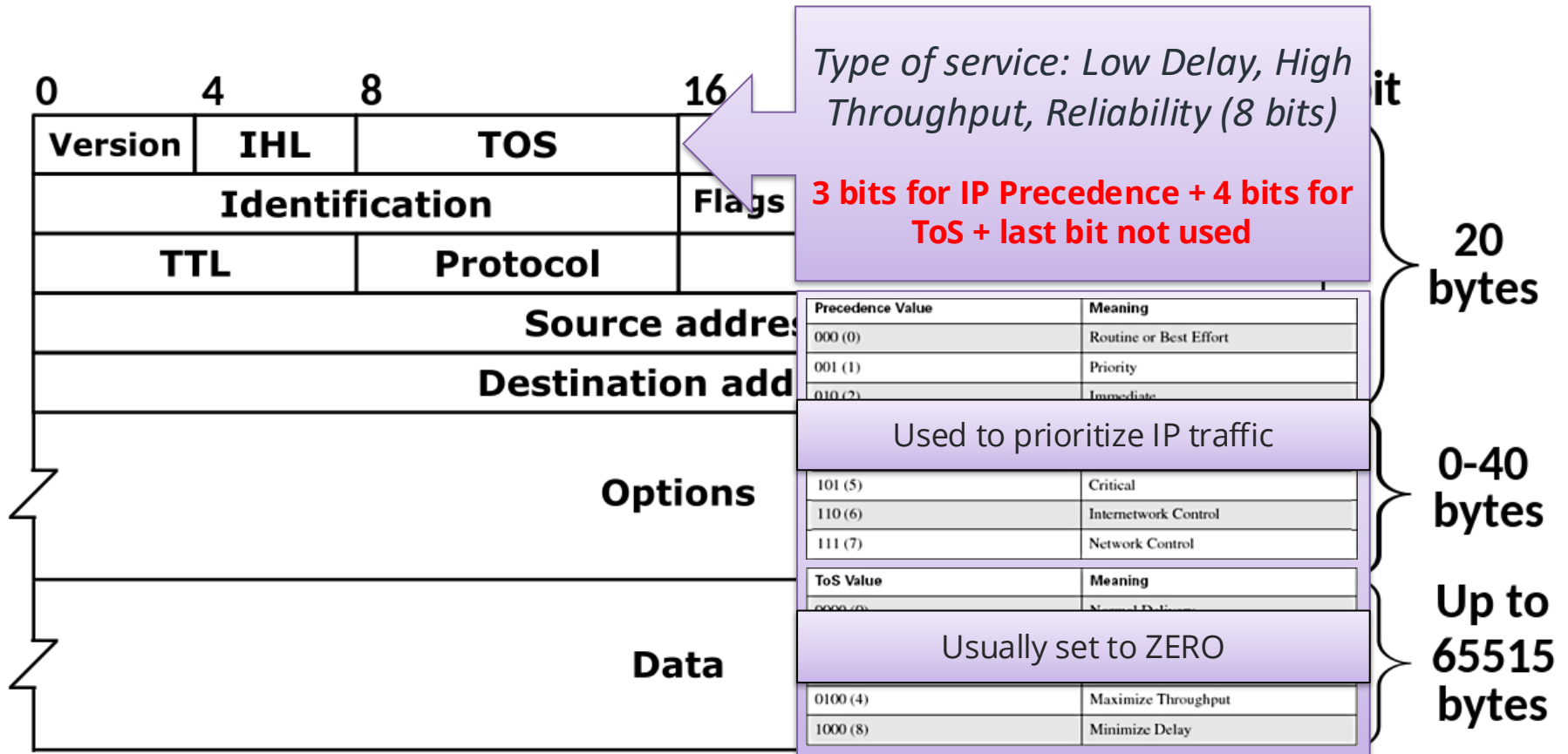
Fourth revision of the Internet Protocol

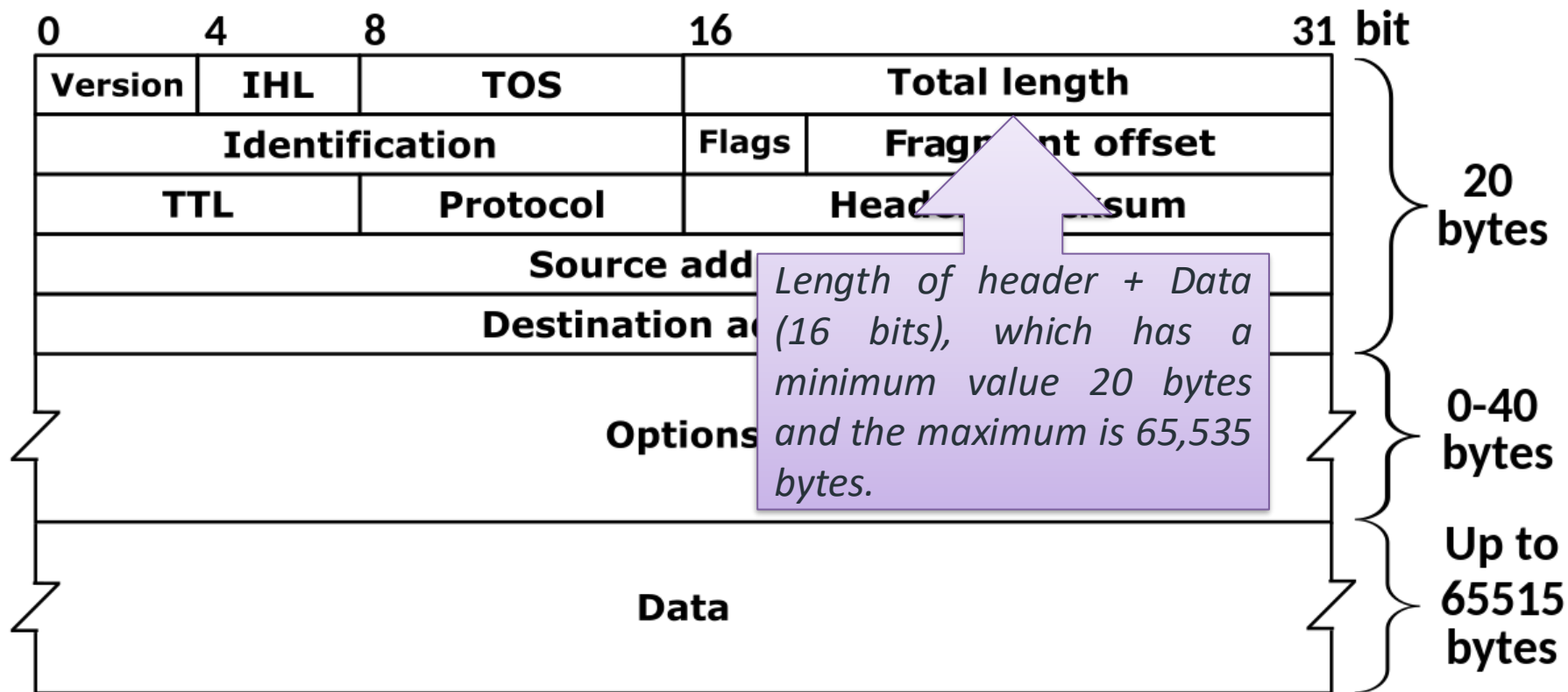


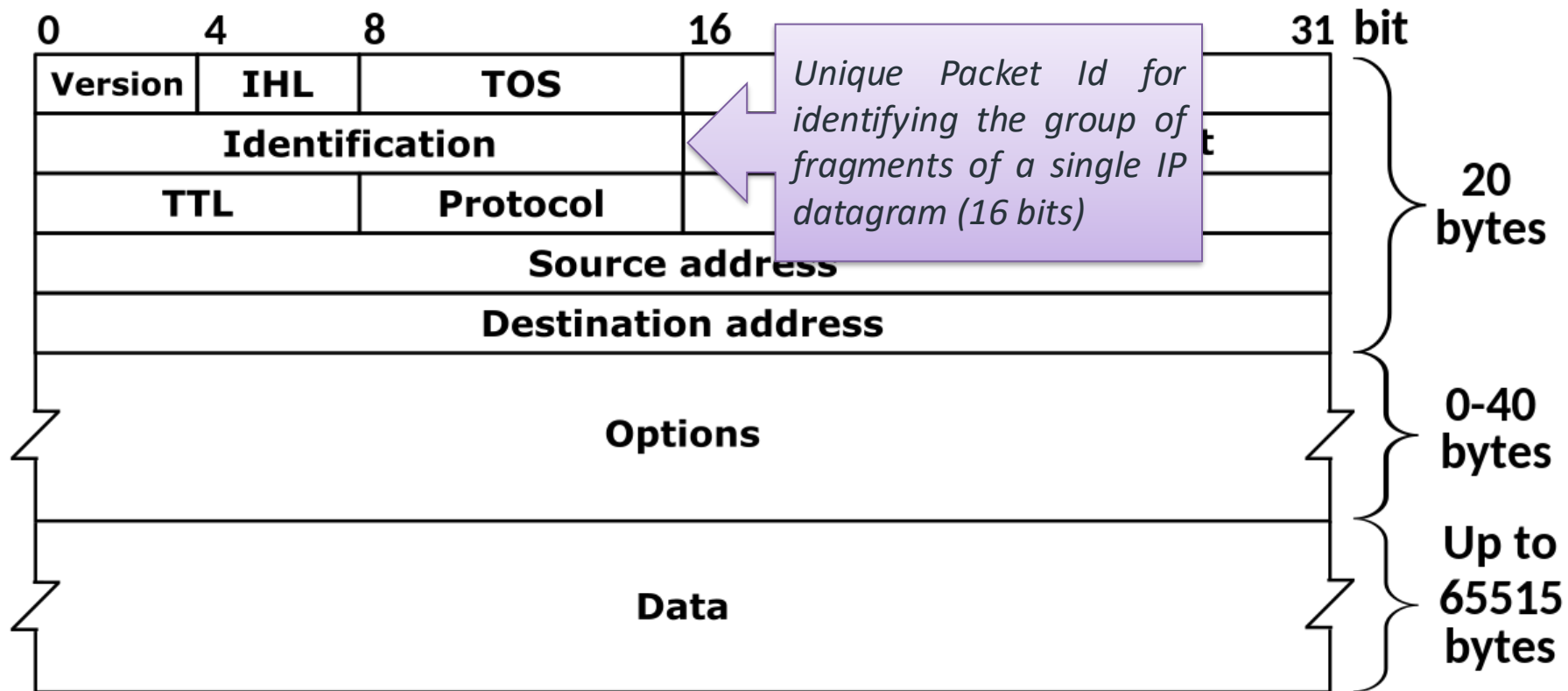


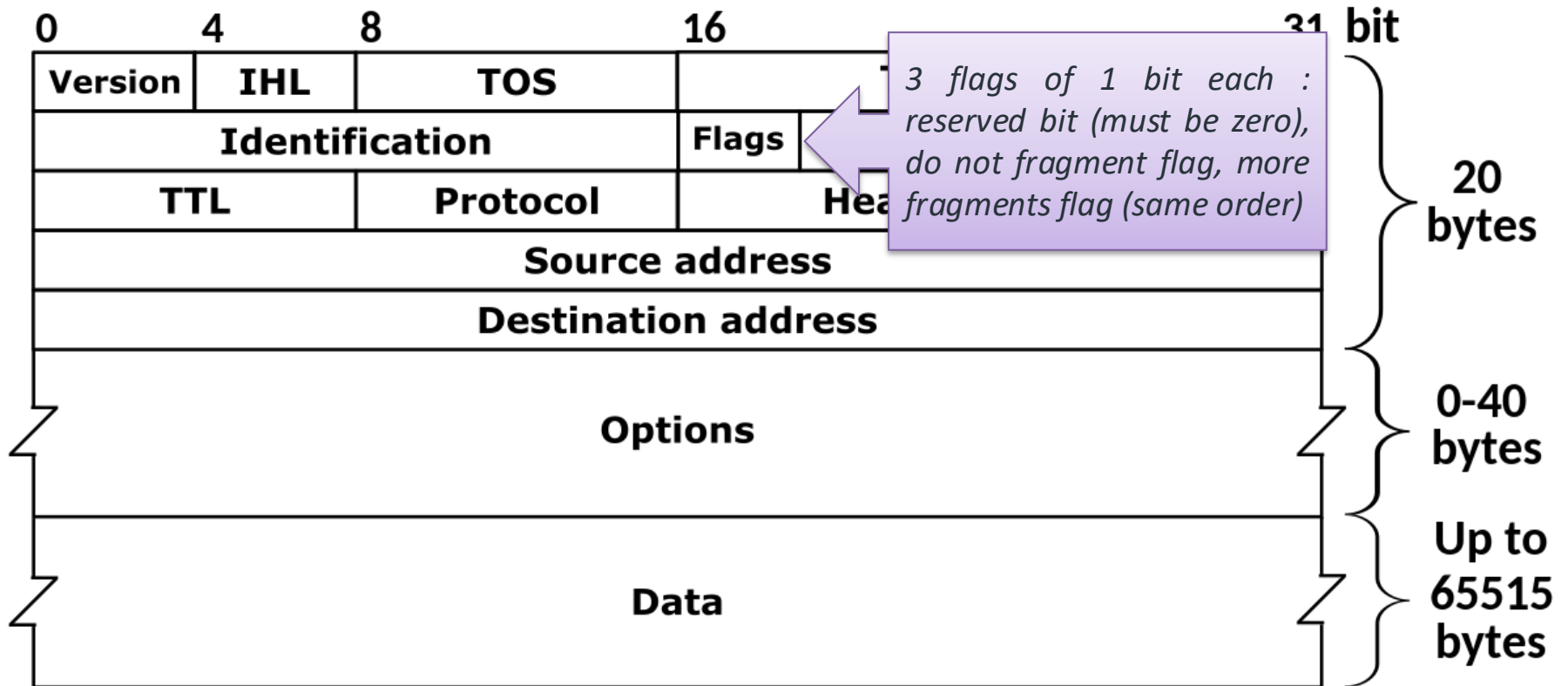


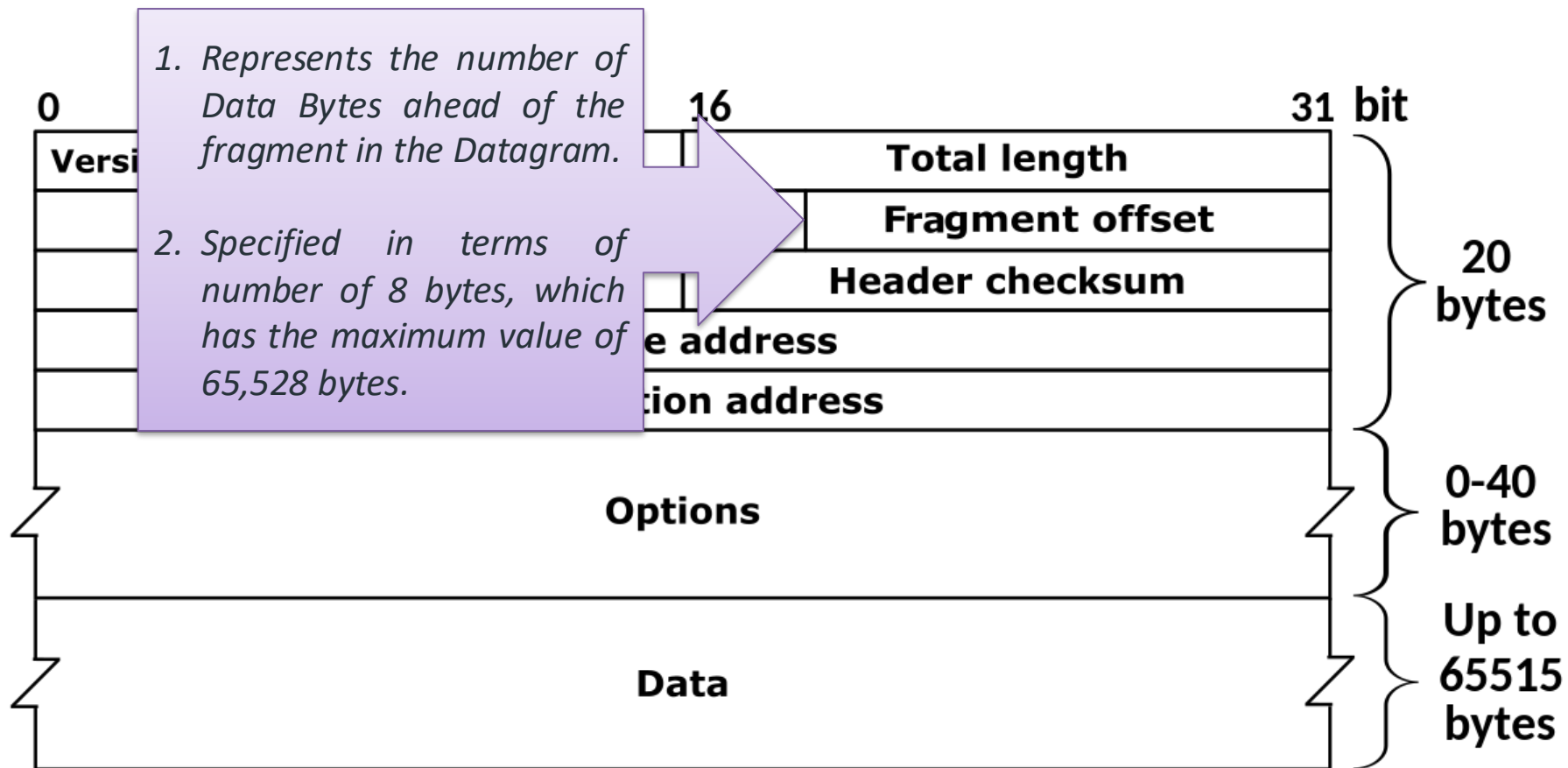












0 4 8 16 31 bit

1. As an IP packet moves through the Internet, it might need to cross a route that cannot handle the size of the packet.
2. The packet will be divided, or fragmented, into smaller packets and reassembled later.
3. These fields are used to fragment and reassemble packets.

Source address

Destination address

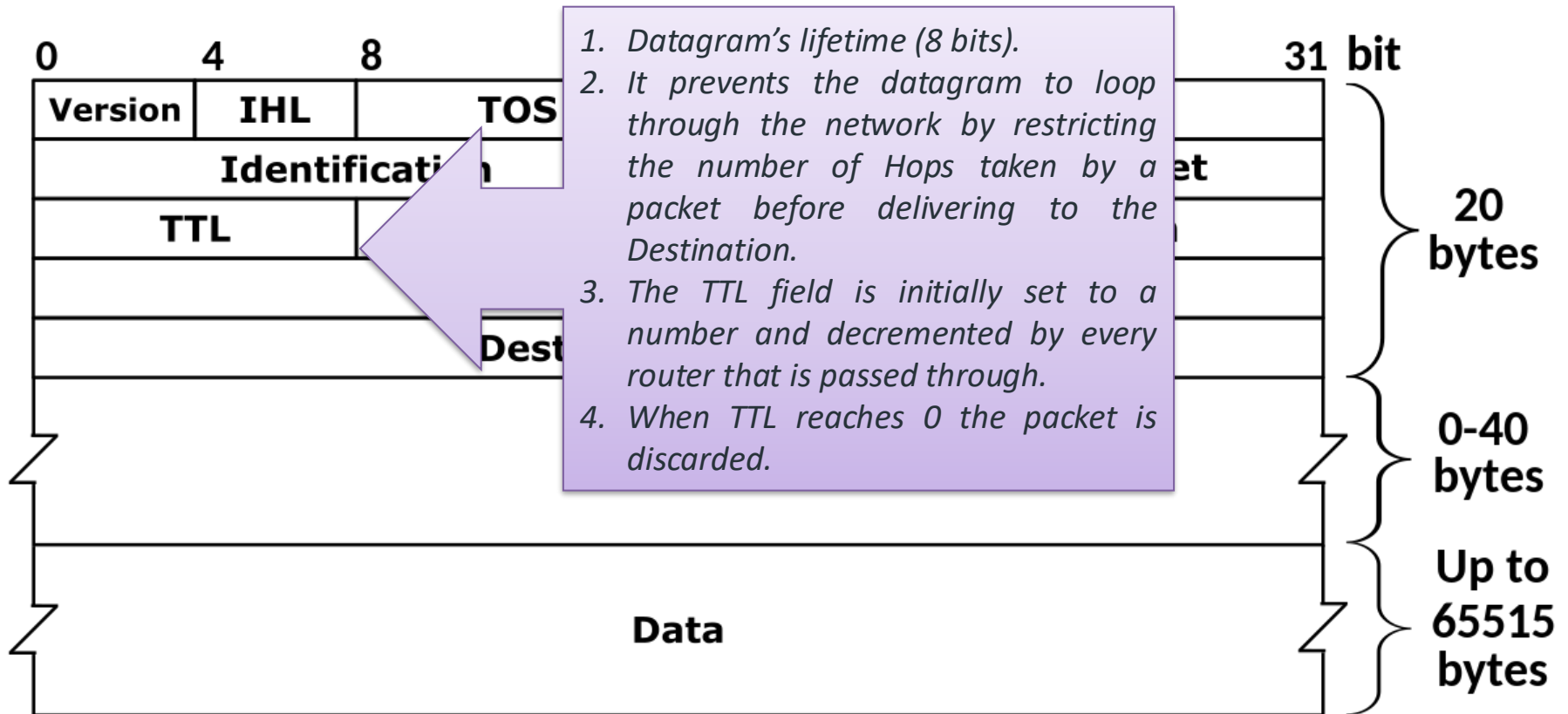
Options

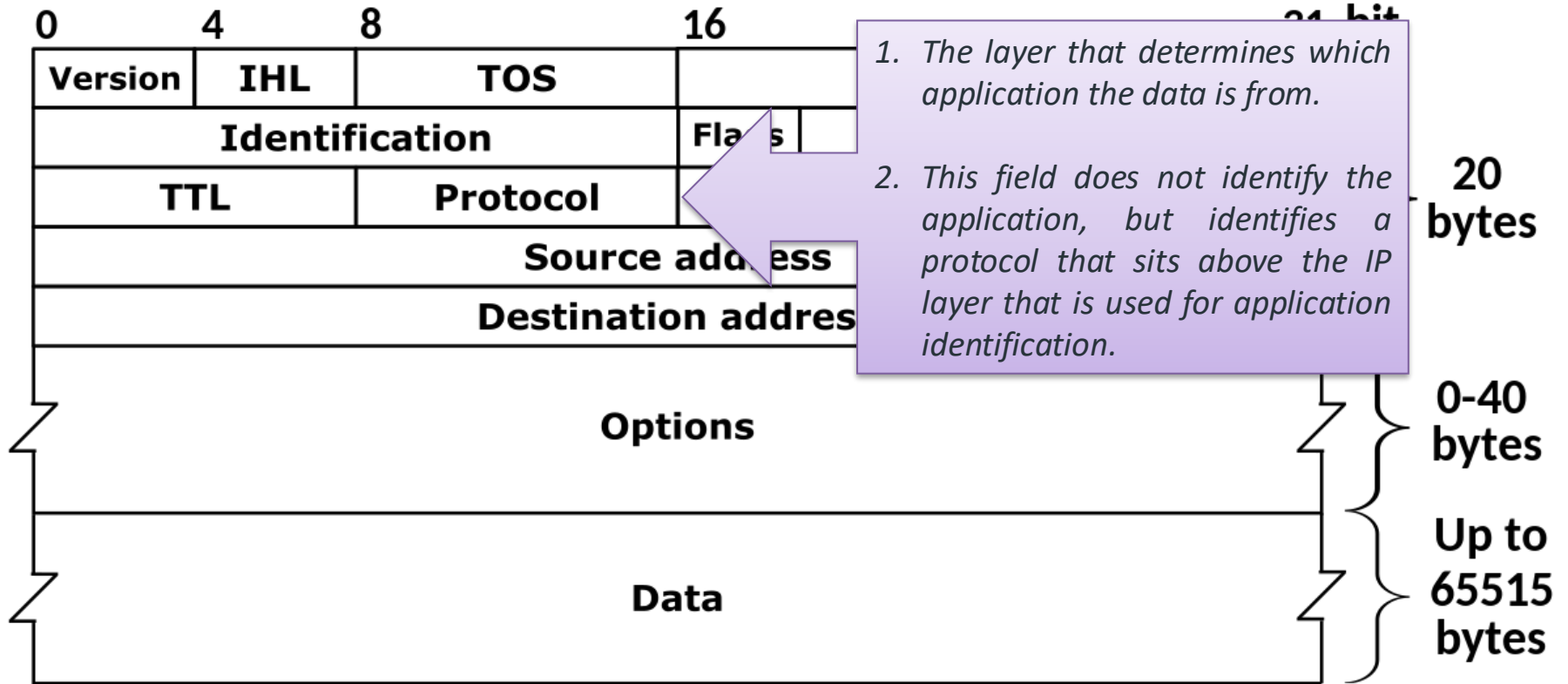
Data

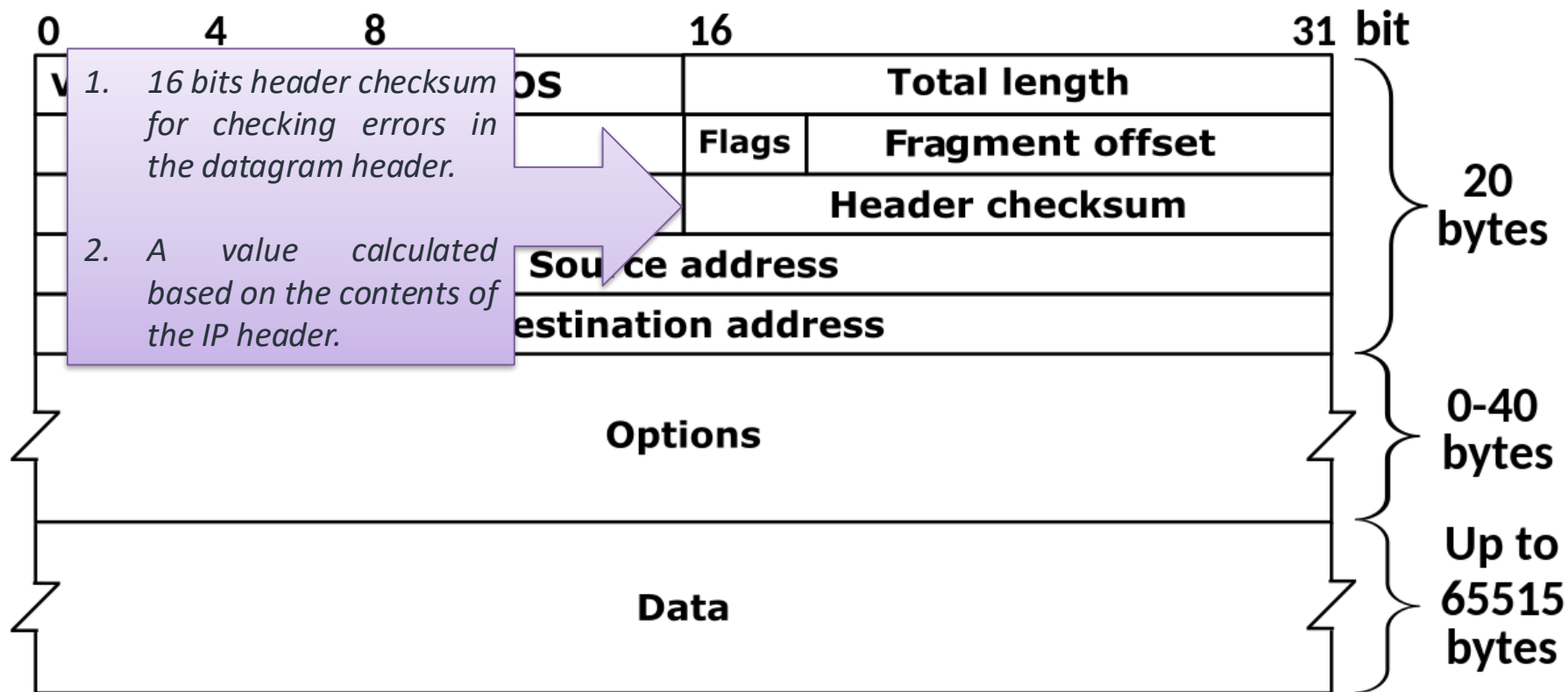
**20
bytes**

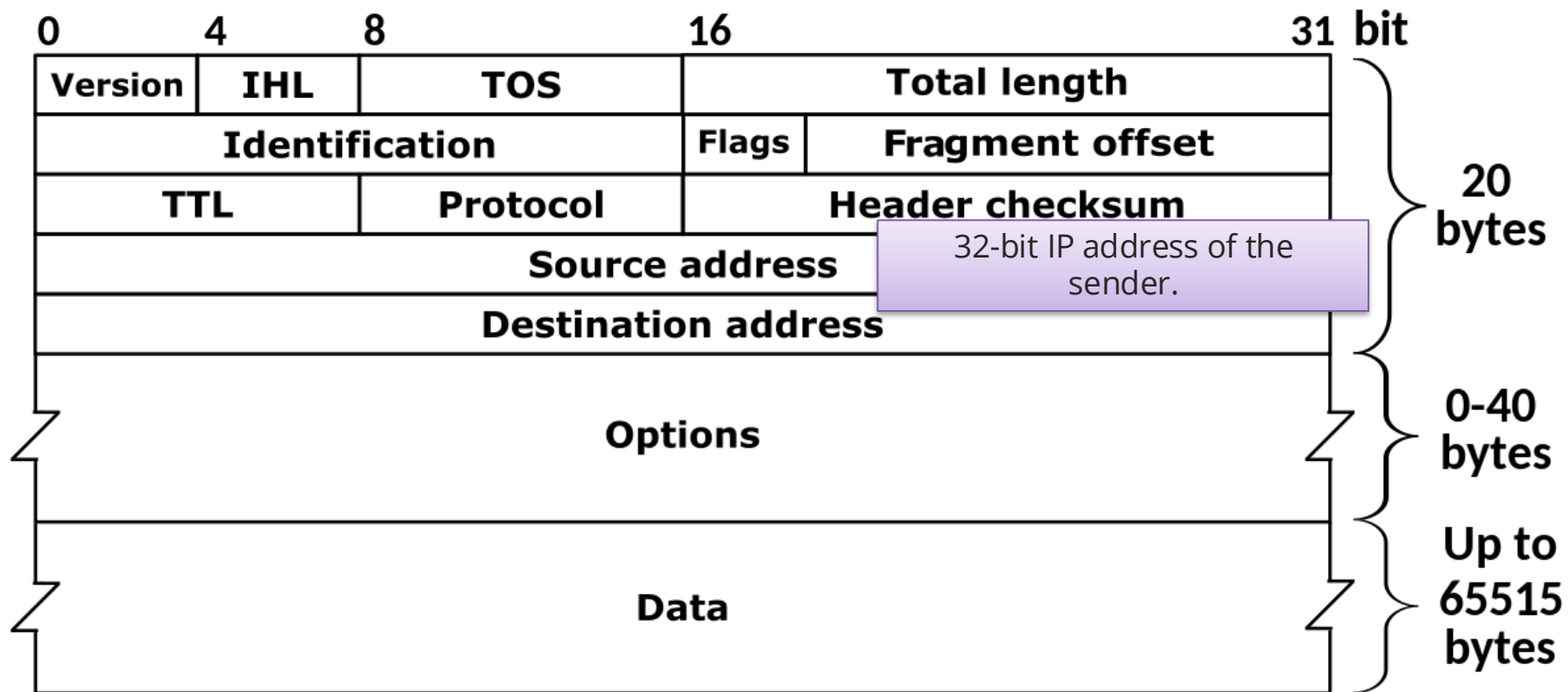
**0-40
bytes**

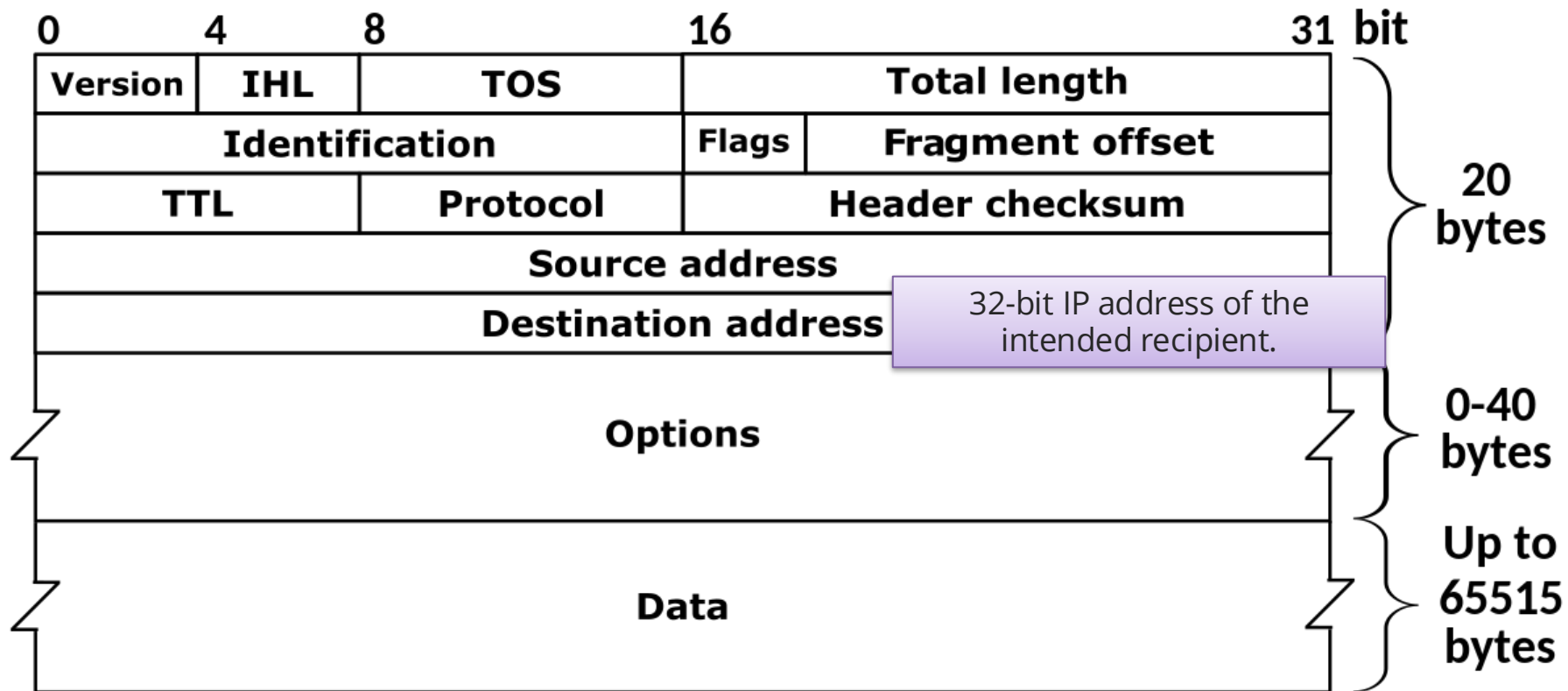
**Up to
65515
bytes**

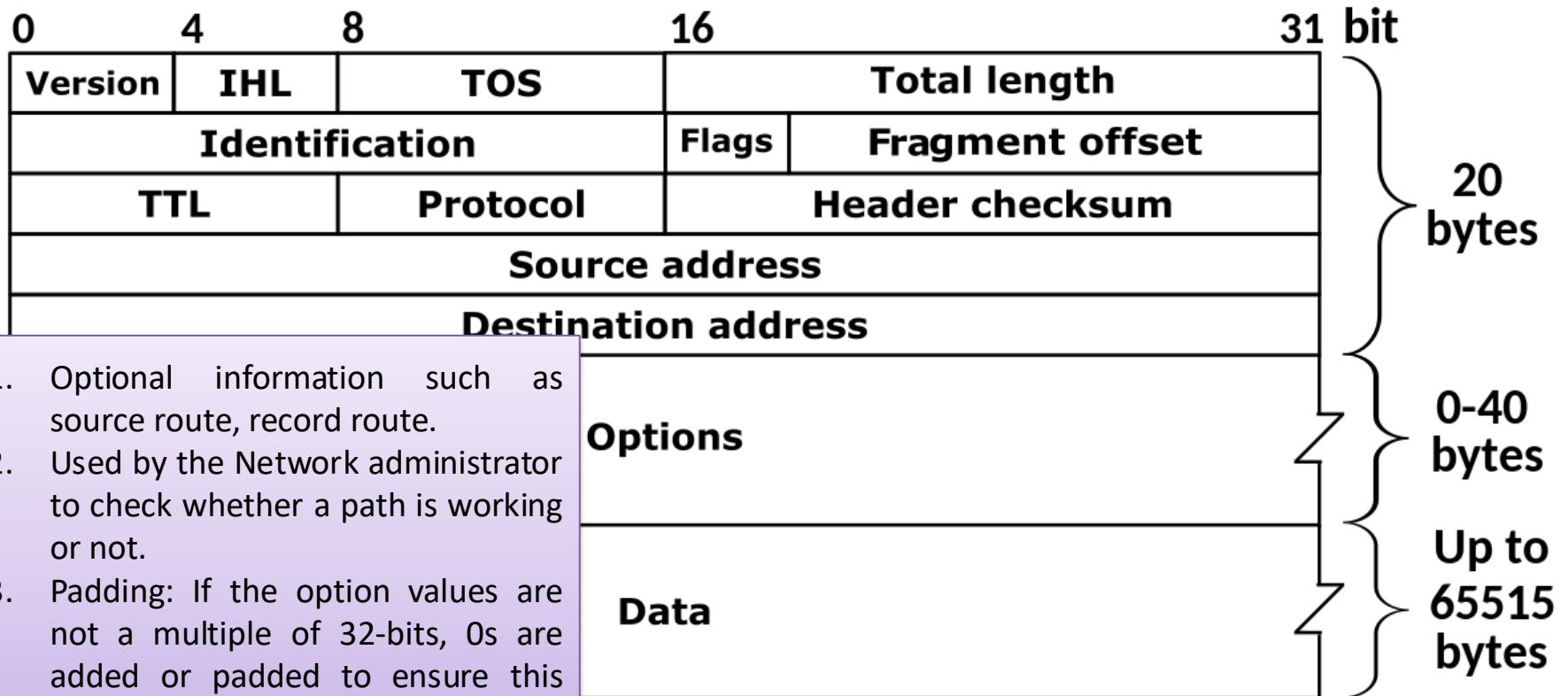




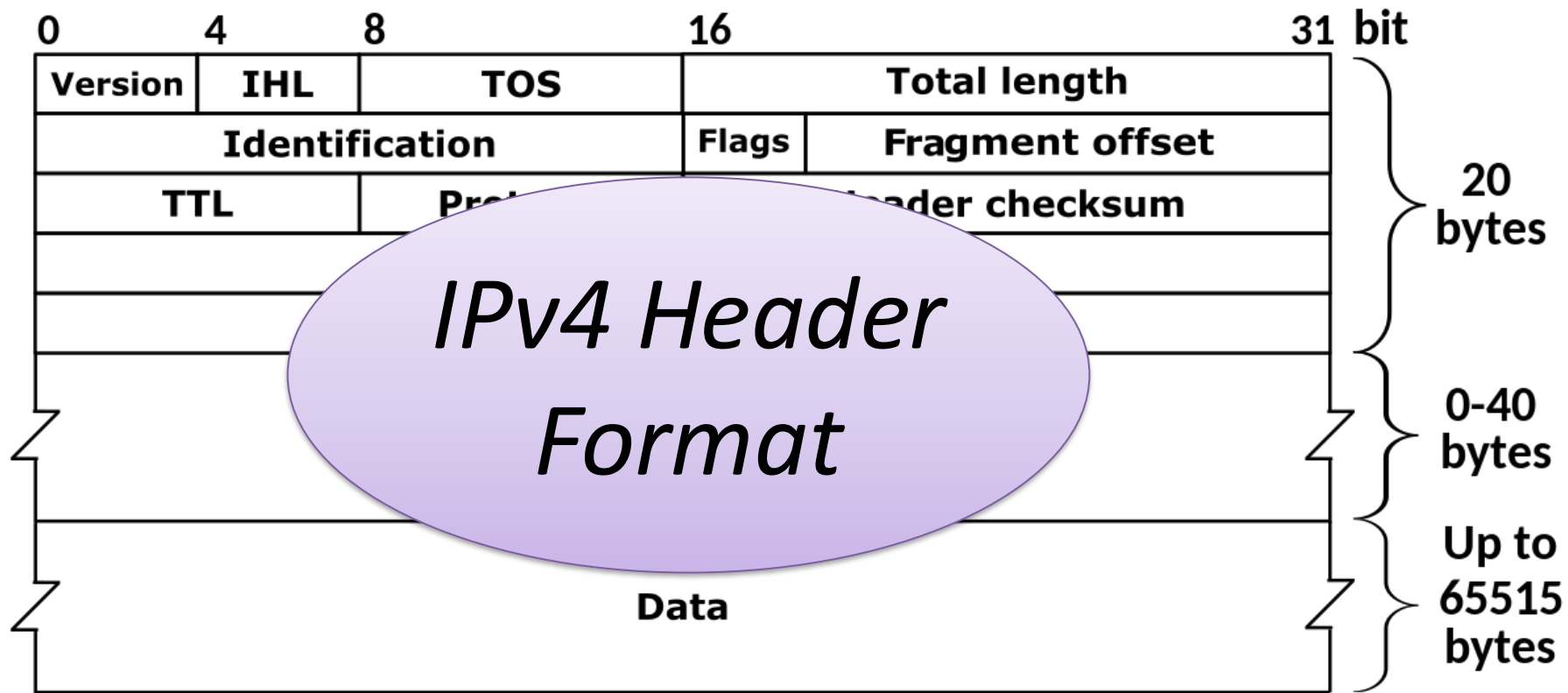








1. Optional information such as source route, record route.
2. Used by the Network administrator to check whether a path is working or not.
3. Padding: If the option values are not a multiple of 32-bits, 0s are added or padded to ensure this field contains a multiple of 32 bits.



IPv6 Header Format

Sixth revision of the Internet Protocol

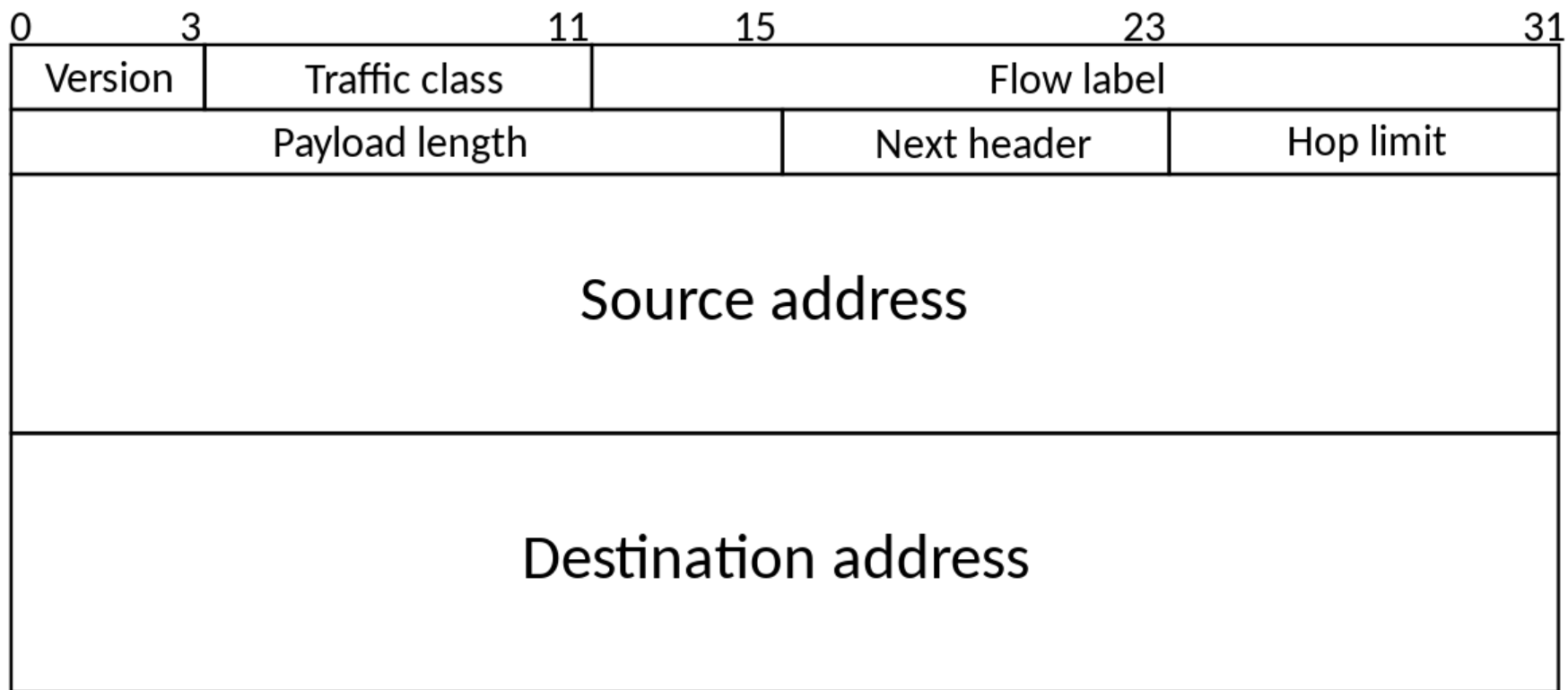
IPv6 addresses are comprised of 8 segments :

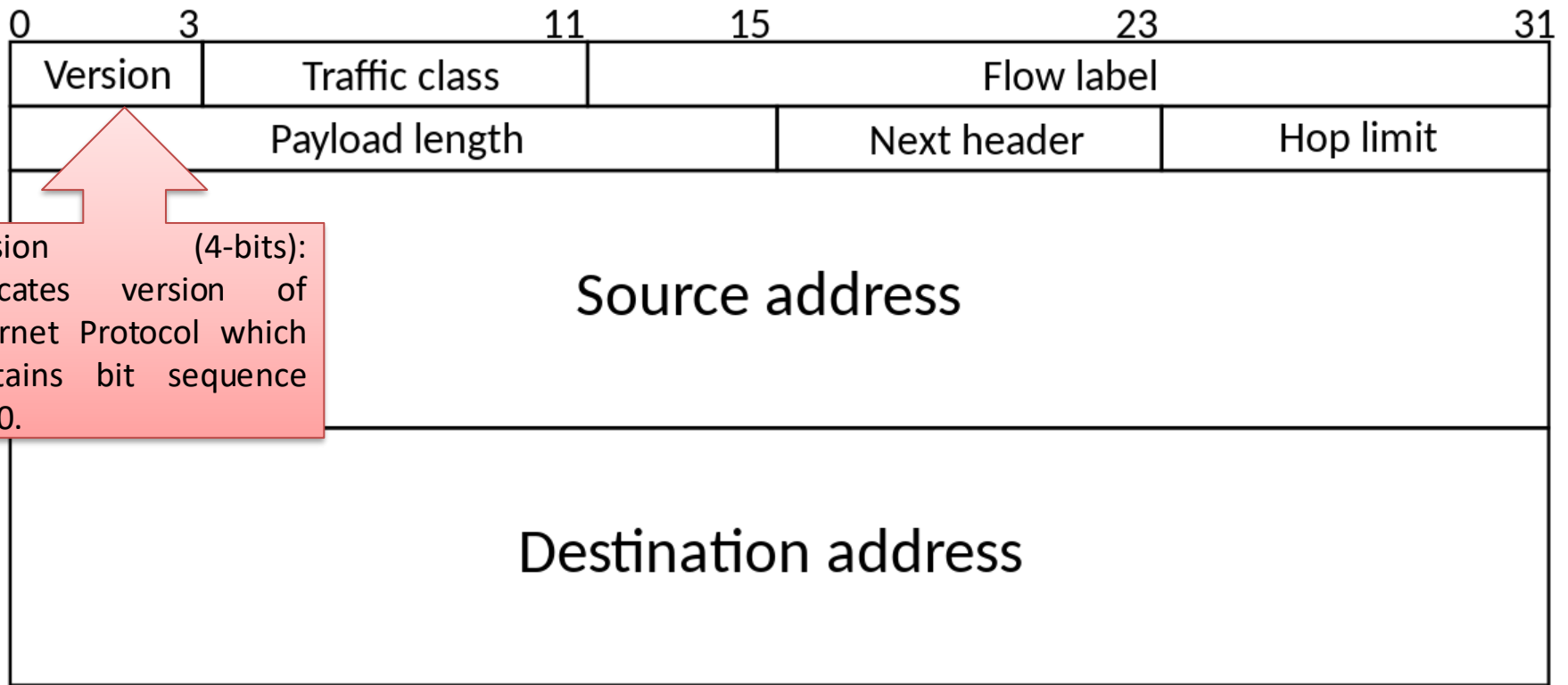
3ffe:1900:4545:3:200:f8ff:fe21:67cf
21DA:D3:0:2F3B:2AA:FF:FE28:9C5A

An IPv6 address is expressed by eight groups of hexadecimal (base-16) numbers separated by colons.

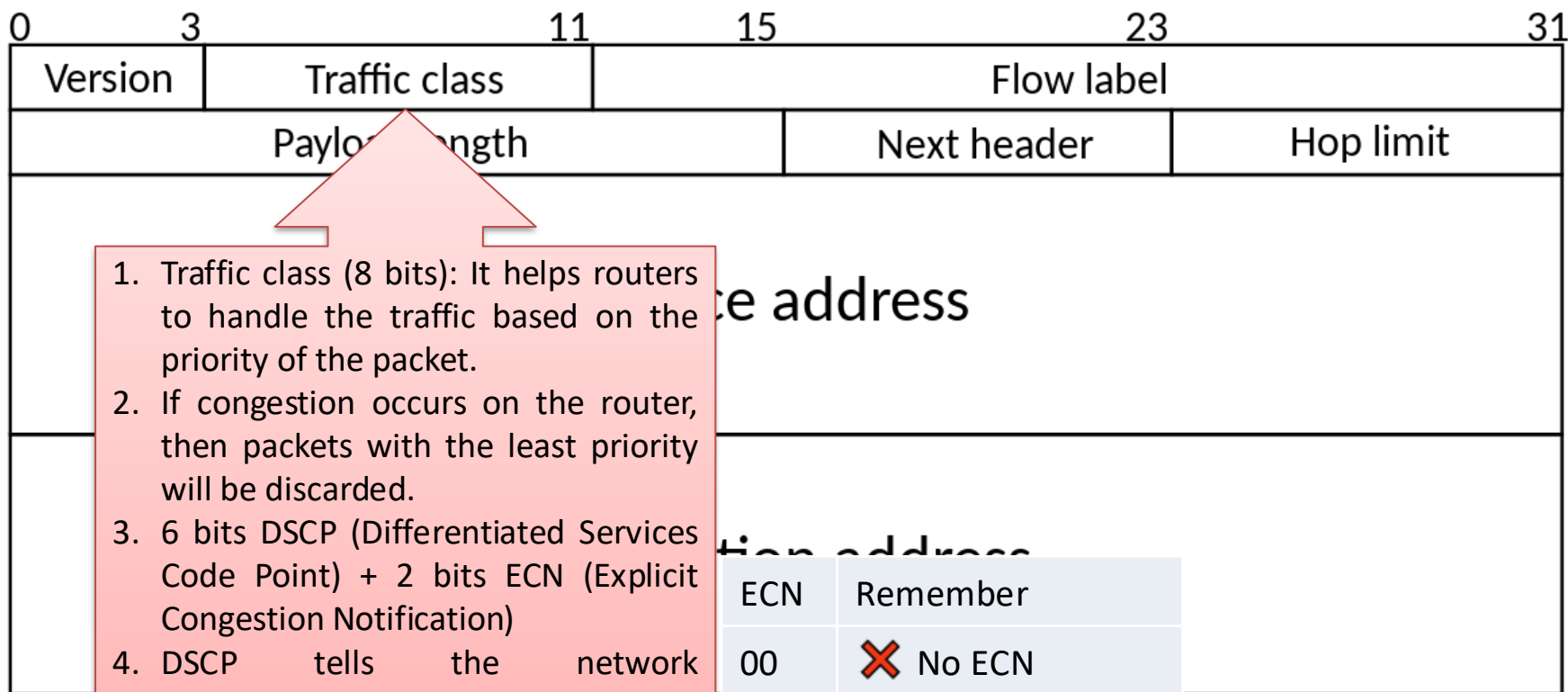
2001:cdba:0000:0000:0000:0000:3257:9652 >>

2001:cdba::3257:9652 [Groups of numbers that contain all zeros are often omitted to save space, leaving a colon separator to mark the gap]

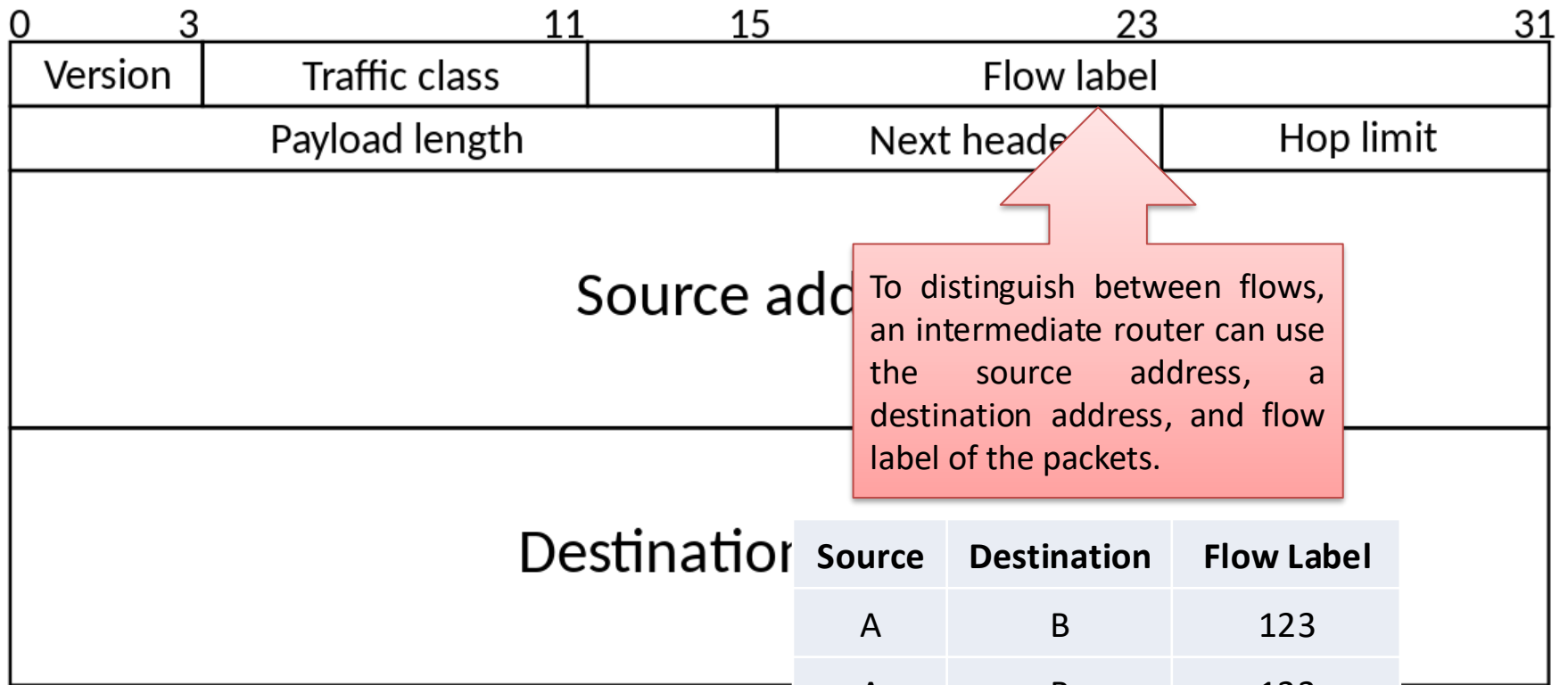




Version (4-bits):
Indicates version of
Internet Protocol which
contains bit sequence
0110.

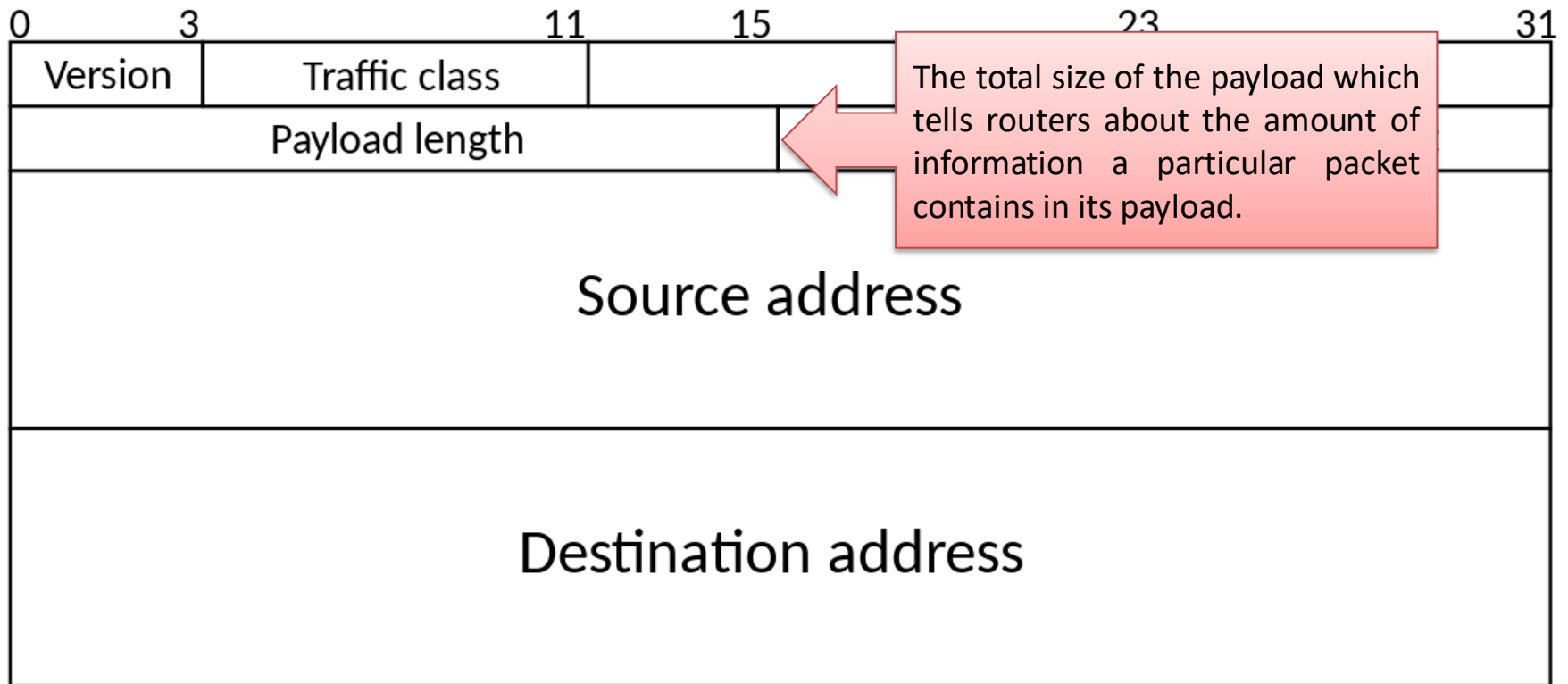


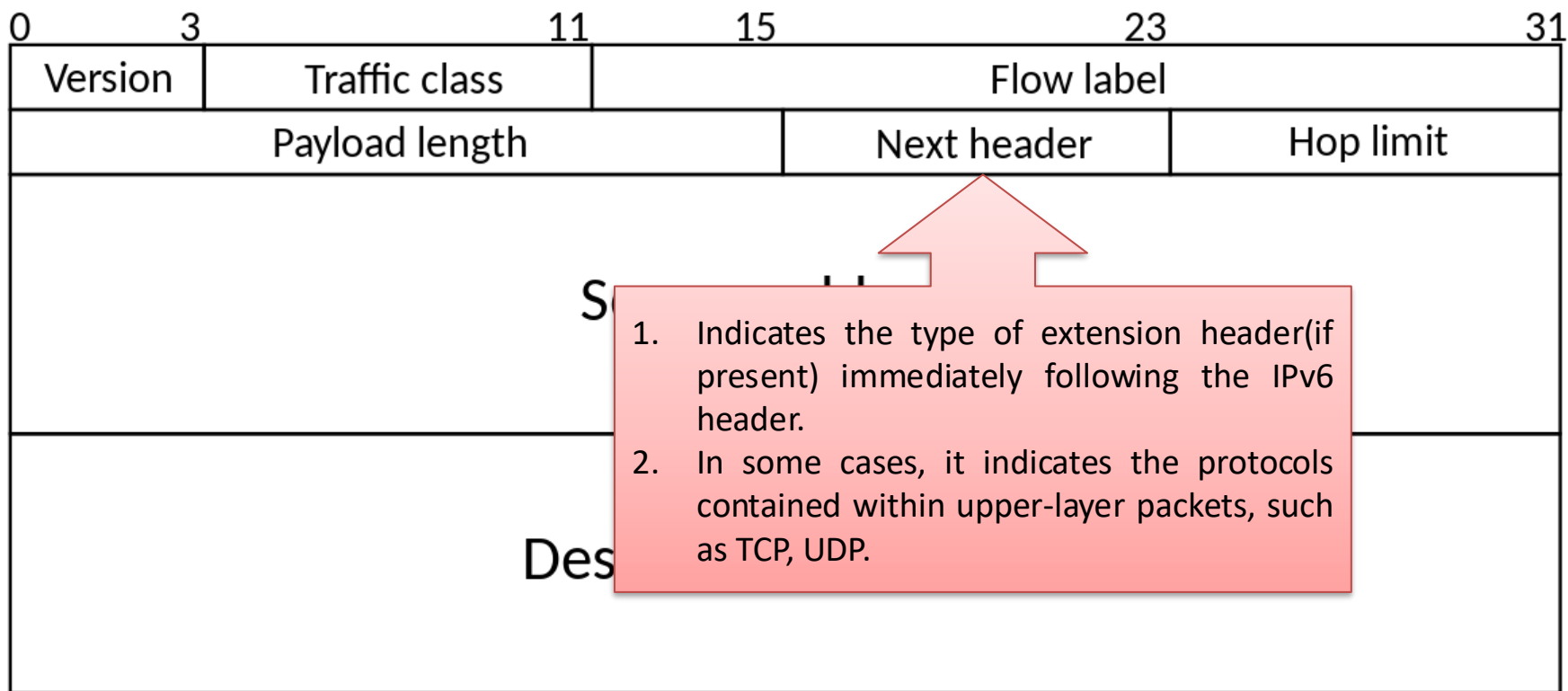
ECN	Remember
00	No ECN
01	ECN capable
10	ECN capable
11	Congestion experienced

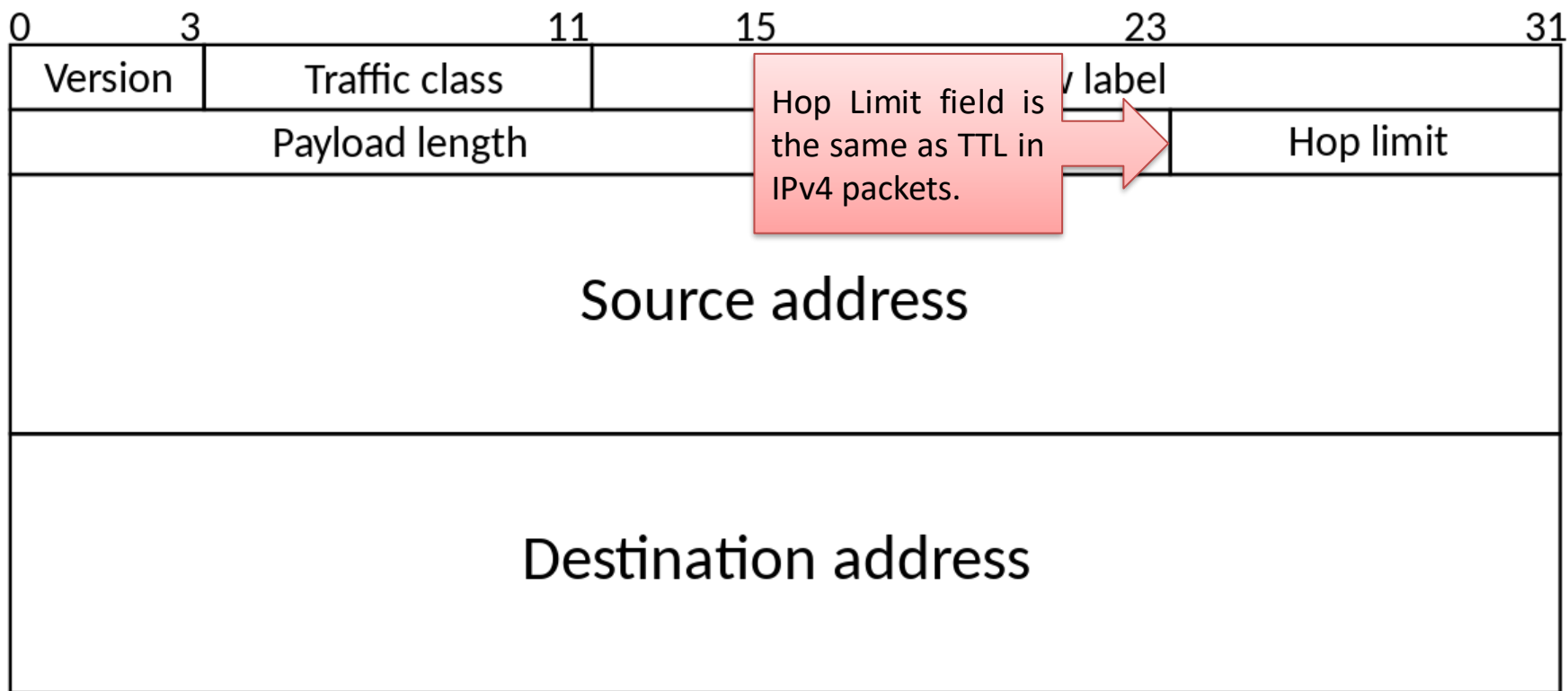


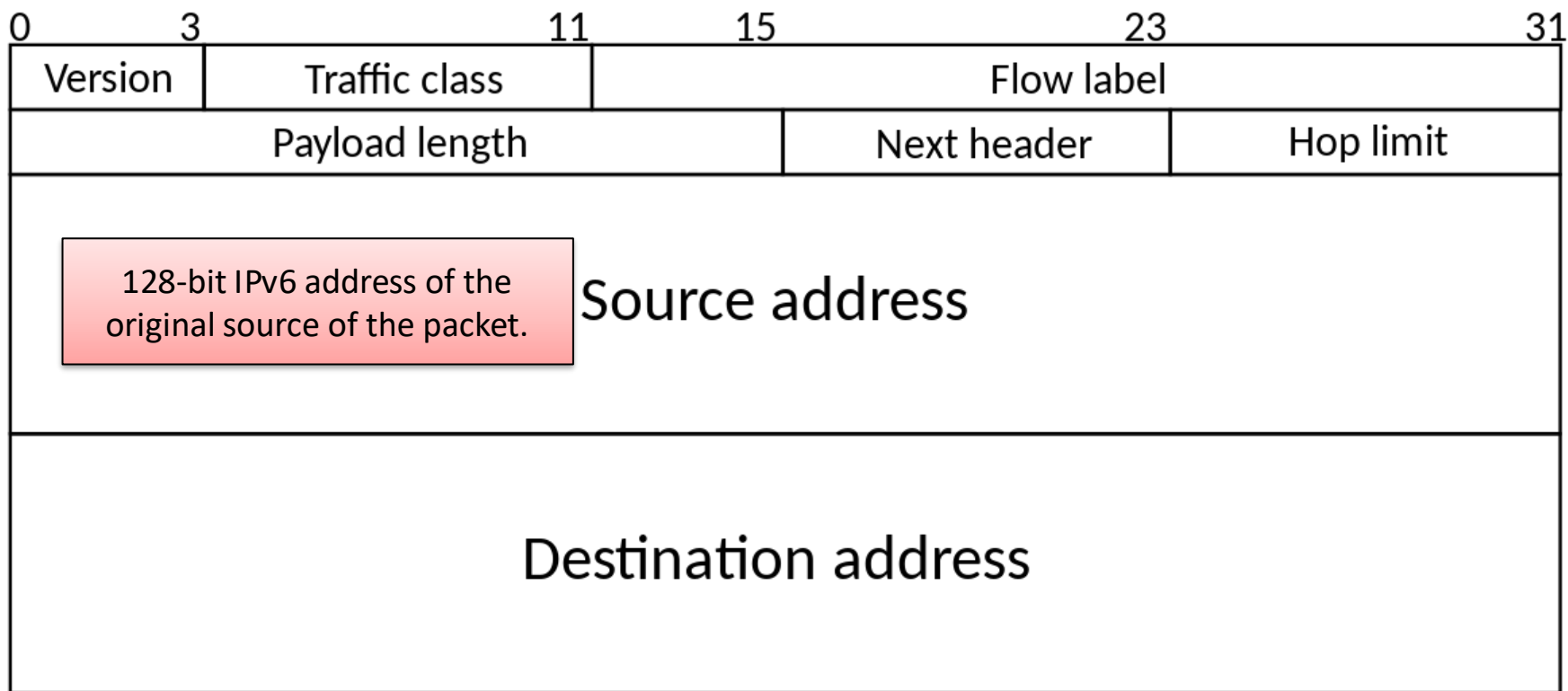
Destination

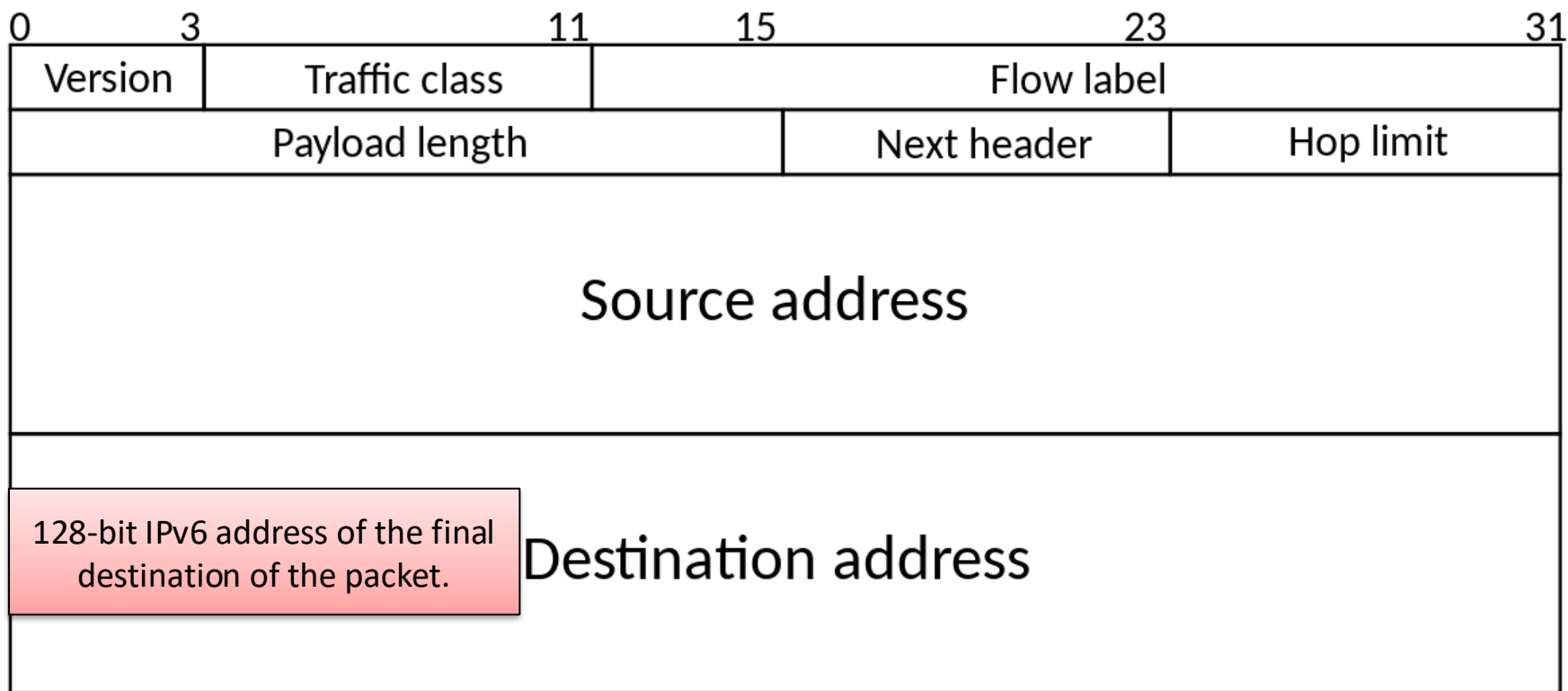
Source	Destination	Flow Label
A	B	123
A	B	123
A	B	456

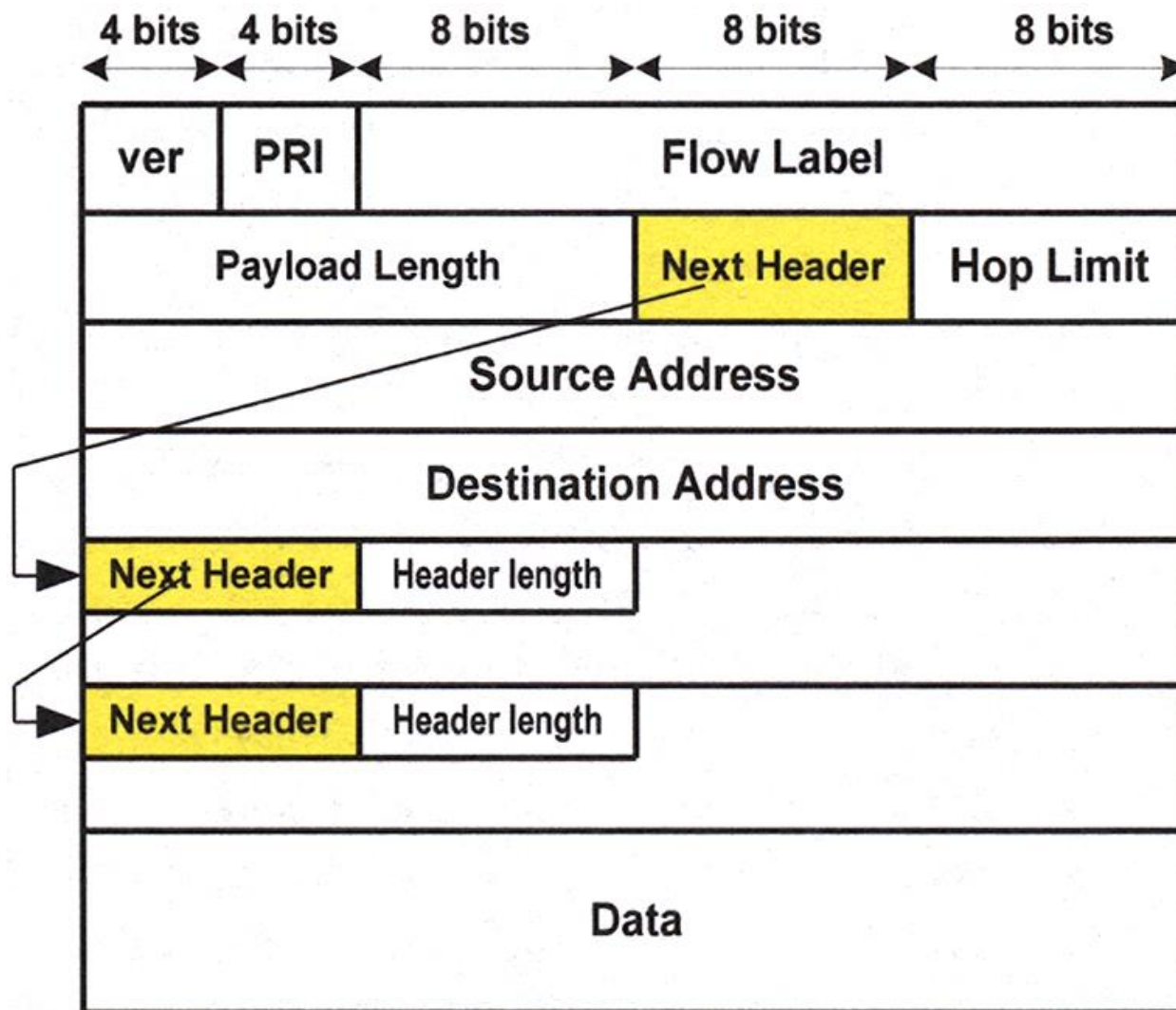






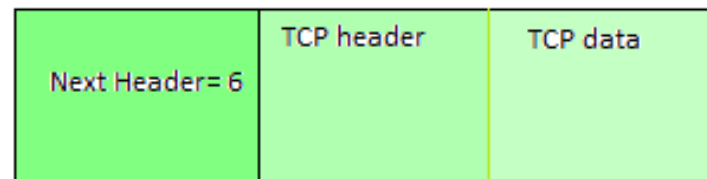




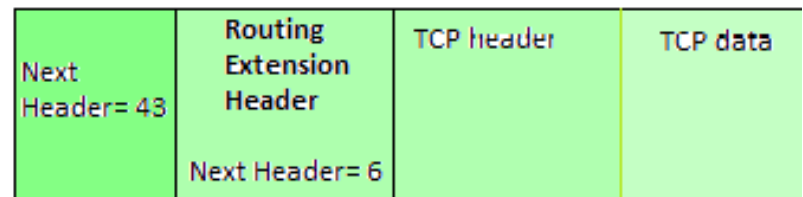


Order	Header Type	Next Header Code
1	Basic IPv6 Header	-
2	Hop-by-Hop Options	0
3	Destination Options (with Routing Options)	60
4	Routing Header	43
5	Fragment Header	44
6	Authentication Header	51
7	Encapsulation Security Payload Header	50
8	Destination Options	60
9	Mobility Header	135
	No next header	59
Upper Layer	TCP	6
Upper Layer	UDP	17
Upper Layer	ICMPv6	58

Example: *TCP is used in IPv6 packet*

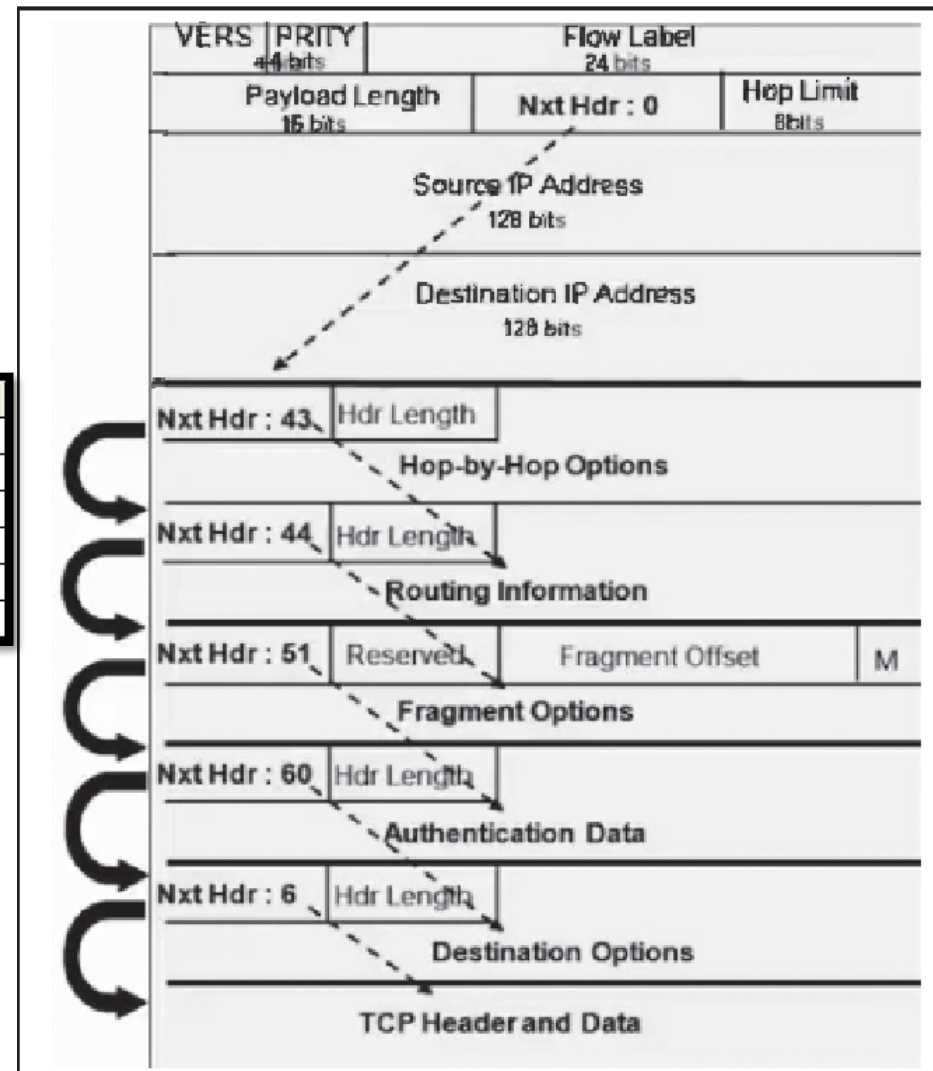


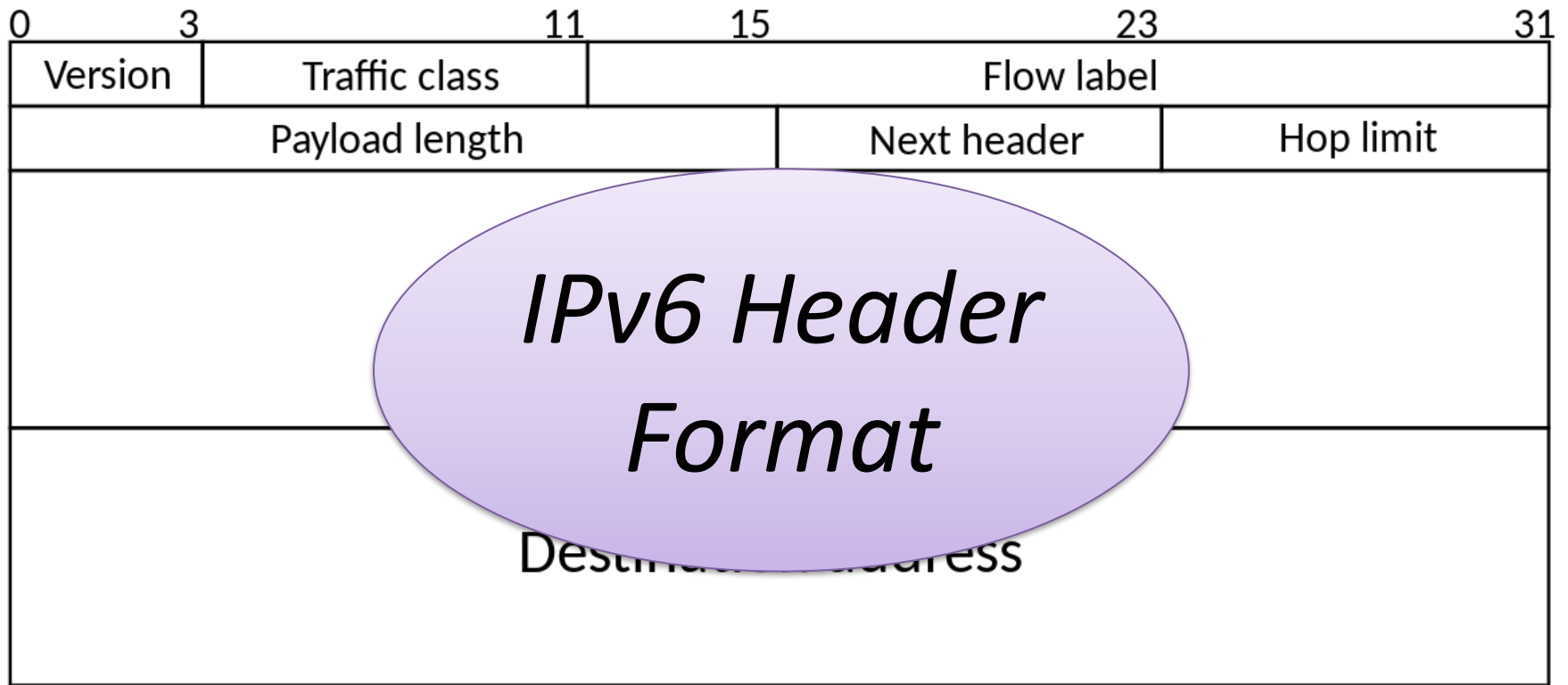
Example2:



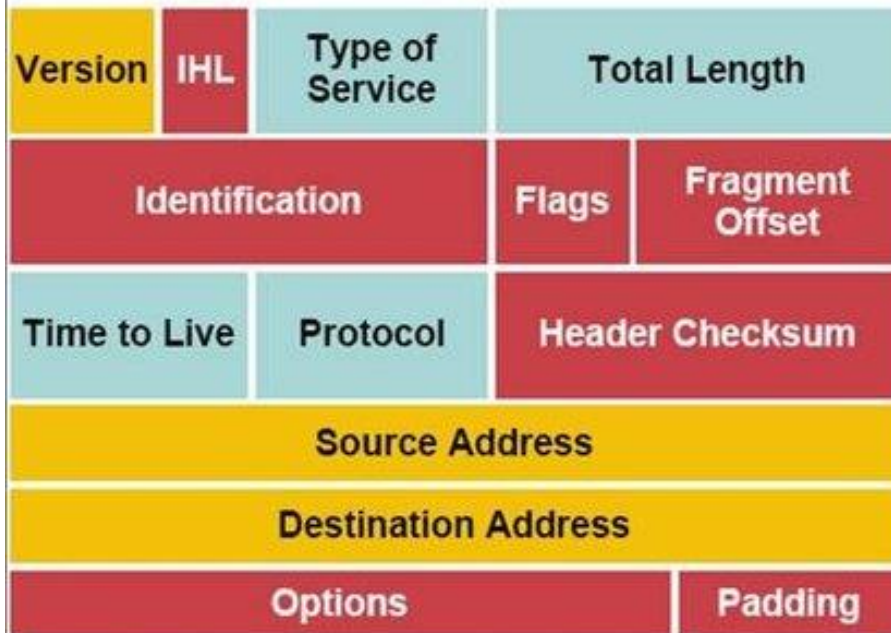
Extension of IPv6 Next Header

Ext. Header	Description
Hop-by-Hop Options	Examined by all devices on the path
Destination Options (with routing options)	Examined by destination of the packet
Routing Header	Methods to take routing decision
Fragment Header	Contains parameters of fragmented datagram done by source
Authentication Header	verify authenticity
Encapsulating Security Payload	Carries Encrypted data





IPv4 Header



Legend

- Field's name kept from IPv4 to IPv6
- Fields not kept in IPv6
- Name and position changed in IPv6
- New field in IPv6

IPv6 Header



IPv4	IPv6
IPv4 has a 32-bit address length	IPv6 has a 128-bit address length
In IPv4 end to end, connection integrity is Unachievable	In IPv6 end to end, connection integrity is Achievable
It can generate 4.29×10^9 address space	Address space of IPv6 is quite large it can produce 3.4×10^{38} address space
The Security feature is dependent on application	IPSEC is an inbuilt security feature in the IPv6 protocol
Address representation of IPv4 is in decimal	Address Representation of IPv6 is in hexadecimal
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation performed only by the sender
In IPv4 Packet flow identification is not available	In IPv6 packet flow identification are Available and uses the flow label field in the header
In IPv4 Encryption and Authentication facility not provided	In IPv6 Encryption and Authentication are provided
IPv4 has a header of 20-60 bytes.	IPv6 has header of 40 bytes fixed
IPv4 consist of 4 fields which are separated by dot (.)	IPv6 consist of 8 fields, which are separated by colon (:)
IPv4's IP addresses are divided into five different classes. Class A , Class B, Class C , Class D , Class E.	IPv6 does not have any classes of IP address.
IPv4 supports VLSM(Variable Length subnet mask).	IPv6 does not support VLSM.
Example of IPv4: 66.94.29.13	Example of IPv6: 2001:0000:3238:DFE1:0063:0000:0000:FEFB
IPv4 uses ARP	IPv6 does not use ARP as it does not support broadcast.

Address Resolution Protocol (ARP)

Introduction

- ARP (Address Resolution Protocol) maps an IPv4 address to the corresponding MAC address on a local network so frames can be delivered to the correct device.
- Works only within the same broadcast domain (LAN)
- Operates directly over link-layer technologies such as Ethernet

Important ARP Terms

1. ARP Cache

- A **temporary storage table** maintained by a device.
- Stores **recently resolved IP-to-MAC address mappings**.
- **Reduces network overhead** by avoiding repeated ARP requests.

2. ARP Cache Timeout

- The time duration for which an entry remains valid in the ARP cache.
- **After expiration**, the entry is **removed** to prevent the use of **outdated mappings**.

Important ARP Terms

3. ARP Request

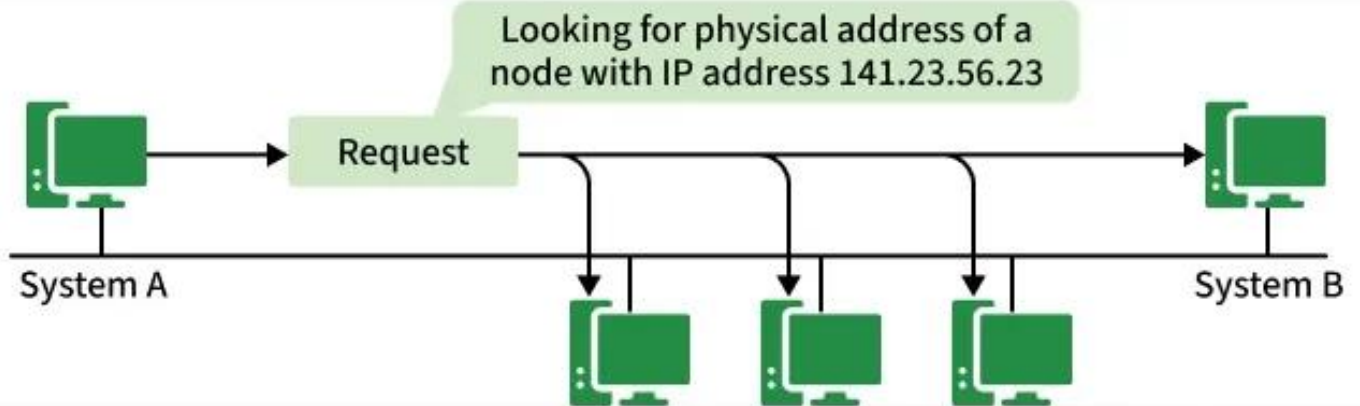
- A **broadcast message** sent to all devices **in the local network**.
- **Asks: “Who has this IP address?”**
- Used **when the MAC address** for a given IP is **not present** in the ARP cache.

4. ARP Reply (ARP Response)

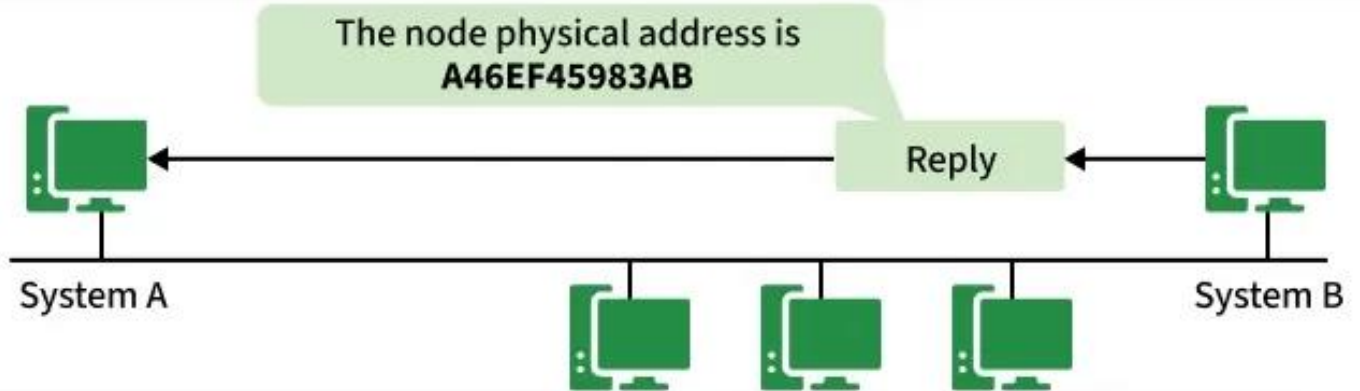
- A **unicast message** sent **by the device that owns the requested IP address**.
- **Contains the corresponding MAC address.**
- Allows the sender to update its ARP cache and proceed with data transmission.

Working

a. ARP request is broadcast



b. ARP reply is unicast



1. ARP Cache Lookup

The sender **first checks its ARP cache** for an existing entry corresponding to the destination IP address.

If a valid entry is found, the MAC address is used immediately and data transmission begins.

2. ARP Request Broadcast

If no cache entry exists, the sender generates an ARP request.

This request is broadcast to all devices on the local network.

3. Request Processing by Hosts

Every device on the LAN receives the ARP request.

Each device compares the requested IP address with its own IP address.

4. ARP Reply Transmission

The device whose IP address matches the request **sends an ARP reply**.

The reply is sent as a unicast message and contains the device's MAC address.

5. ARP Cache Update

Upon receiving the reply, the **sender stores the IP–MAC mapping in its ARP cache.**

This cached entry is used for future communication **until it expires**.

Dynamic Host Configuration Protocol (DHCP)

Introduction

Seamless network connectivity is achieved by **automatically assigning IP addresses and configuration settings** to devices using the Dynamic Host Configuration Protocol (DHCP).

1. **Assigns a unique IP address** to each device.
2. **Provides the subnet mask** to define the local network range.
3. **Configures the default gateway** for communication outside the local network.
4. **Supplies DNS server addresses** for domain name resolution.

Components

DHCP Server: The device (router/server) that manages the DHCP service, **keeps the IP pool, and assigns IP addresses + configuration** to clients.

DHCP Relay: A **router/switch** feature that **forwards DHCP requests** and replies between clients and the server when they are in different networks (different subnets).

DHCP Client: **Any device** (PC, mobile, printer, IoT) that requests an IP address and settings from the DHCP server.

IP Address Pool: The **range of IP addresses available for allocation to clients** (often with exclusions/reservations).

Components

Lease: The time period for which an assigned IP address is valid; the **client must renew the lease to keep using it.**

DHCP Options: **Additional configuration parameters** provided by the DHCP server along with the IP address, such as **DNS servers, domain name, and other network-specific settings.**

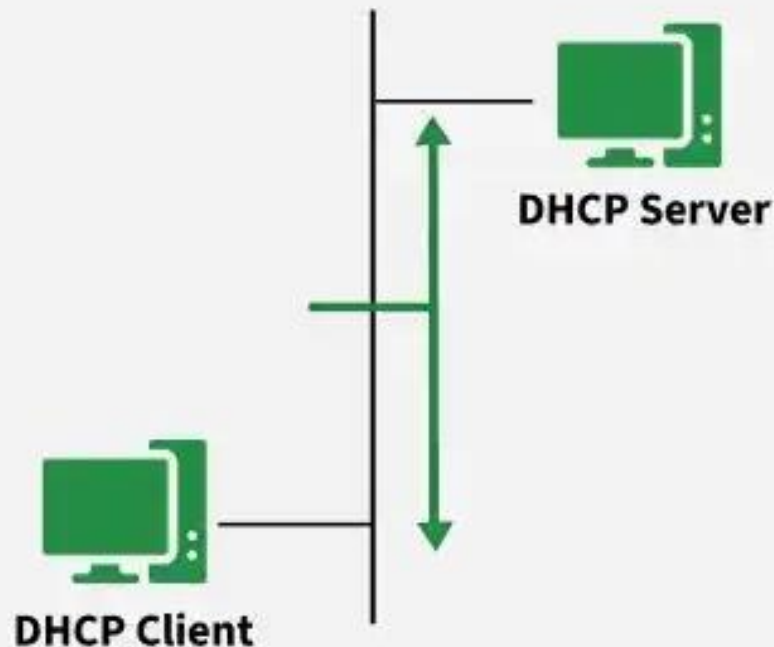
Default Gateway: The **router IP given to the client** so it can communicate outside the local network (reach the internet/other networks).

DNS Servers: **DNS addresses provided to the client** so it can convert domain names (like google.com) into IP addresses.

Working

1. DHCP Discover Message

- **Triggered when:** A device connects to a network or boots up with no valid IP configuration.
- **Transmission type:** Broadcast, because the client does not know the DHCP server's address.
- **Purpose:** Identify DHCP server(s) capable of assigning an IP and network parameters.



DHCP DISCOVER

Dest MAC Address: FFFFFFFFFFFFFFFF

Source MAC Addr: 08002B3EAF2A

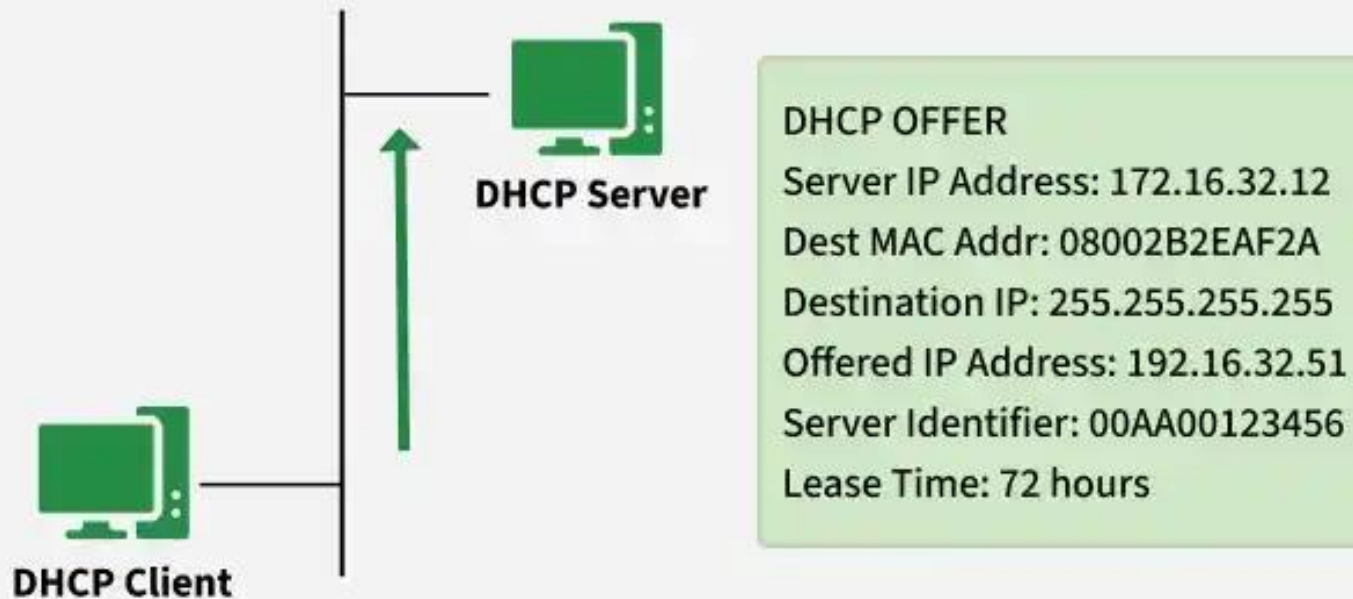
Source IP Address: 0.0.0.0

Dest IP Address: 255.255.255.255

Client Identifier: 08002B2EAF2A

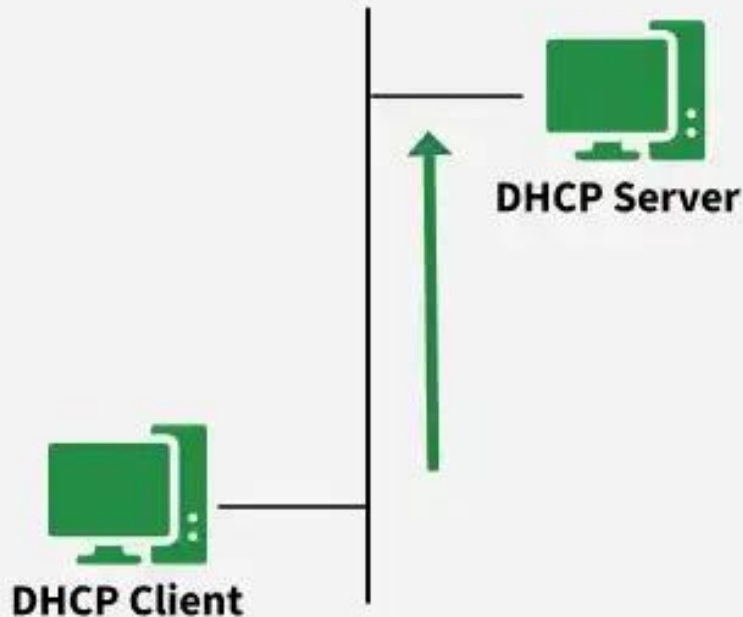
2. DHCP Offer Message

- Can be broadcast or unicast (depends on client/network behaviour)
- Includes Server Identifier so the client knows which server sent the offer
- The offered IP is carried in yiaddr (“your IP address” field)



3. DHCP Request Message

- **Sent by:** DHCP client
- **When it is sent:** After the client receives one or more DHCP Offer messages
- It accepts one offered IP address and identify the selected server
- Prevent multiple servers from assigning different IPs to the same client

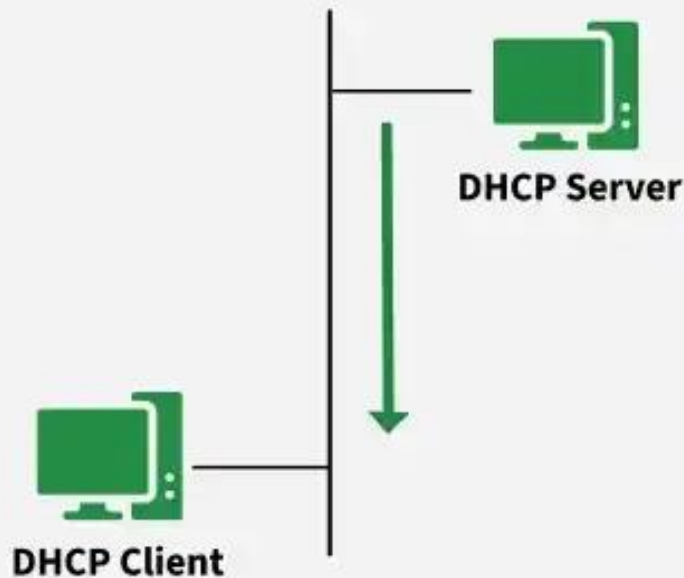


DHCP REQUEST

Dest MAC Address: FF:FF:FF:FF:FF:FF
Source IP Address: 0.0.0.0.
Dest IP Address: 255.255.255.255
Source MAC Addr: 08002B2EAF2A
Server Identifier: 172.16.32.12
Request IP Address: 172.16.32.51
Client Identifier: 08002B2EAF2A
Request parameters.....

4. DHCP Acknowledgment Message

- **Sent by:** DHCP server
- **When it is sent:** After the server accepts the client's DHCP Request
- **Purpose:** Confirm the lease and deliver final configuration so the client can start network communication.



DHCP ACK
Dest MAC Add: 00:11:22:33:44:55 (client's MAC)
Source MAC Addr: 00AA00123456
Source IP Address: 172.16.32.12
Dest IP Address: 255.255.255.255
IP Address: 172.16.32.51
Server Identifier: 172.16.32.12
lease Length: 72 Hours
Client Identifier: 08002B2EAF2A
Other Request Paraments.....

5. DHCP Negative Acknowledgment Message

- **Sent by:** DHCP server
- **Purpose:** Reject the client's DHCP Request
- Requested IP is outside the configured scope
- IP is already allocated to another device
- Client has moved to a different network
- Client must restart the DHCP process from Discover

6. DHCP Decline

- **Sent by:** DHCP client
- **Purpose:** Inform the server that the offered IP is not safe to use
- **Typical trigger:** Client detects an IP conflict using an ARP probe / gratuitous ARP
- **Result:** Server marks the IP as unavailable and selects another address for future allocation.

7. DHCP Release

- **Sent by:** DHCP client
- **Purpose:** Free the IP for reuse
- Occurs when a device disconnects, shuts down, or disables the interface
- Helps maintain efficient IP address utilization

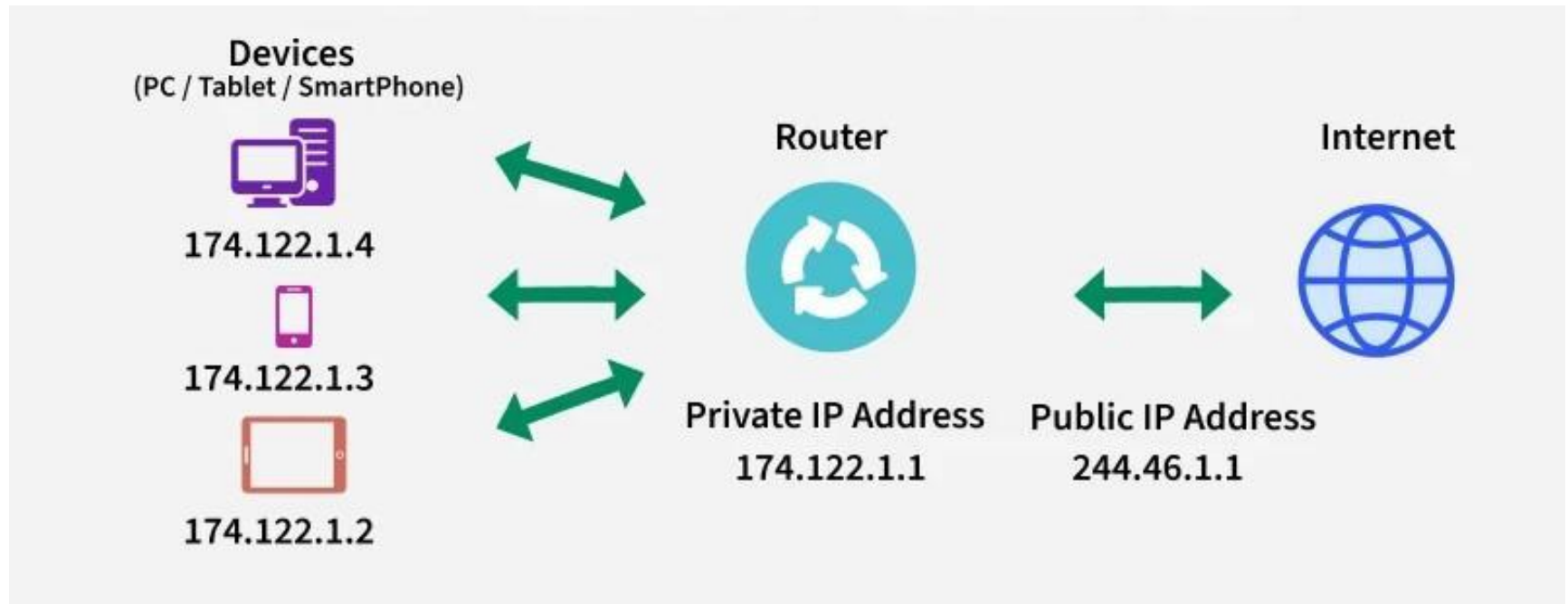
8. DHCP Inform

- **Sent by:** DHCP client
- **Purpose:** Request configuration parameters like DNS, domain name, NTP, etc., without obtaining a new IP.
- **Server response:** Sends a DHCP ACK containing only configuration options.
- **Delivery:** Typically **unicast**, since the client already has a valid IP.

Network Address Translation (NAT)

Introduction

- Network Address Translation (NAT) allows multiple devices in a private network to access the internet using a single public IP address.
- It helps conserve IPv4 addresses and hides internal systems for added security.

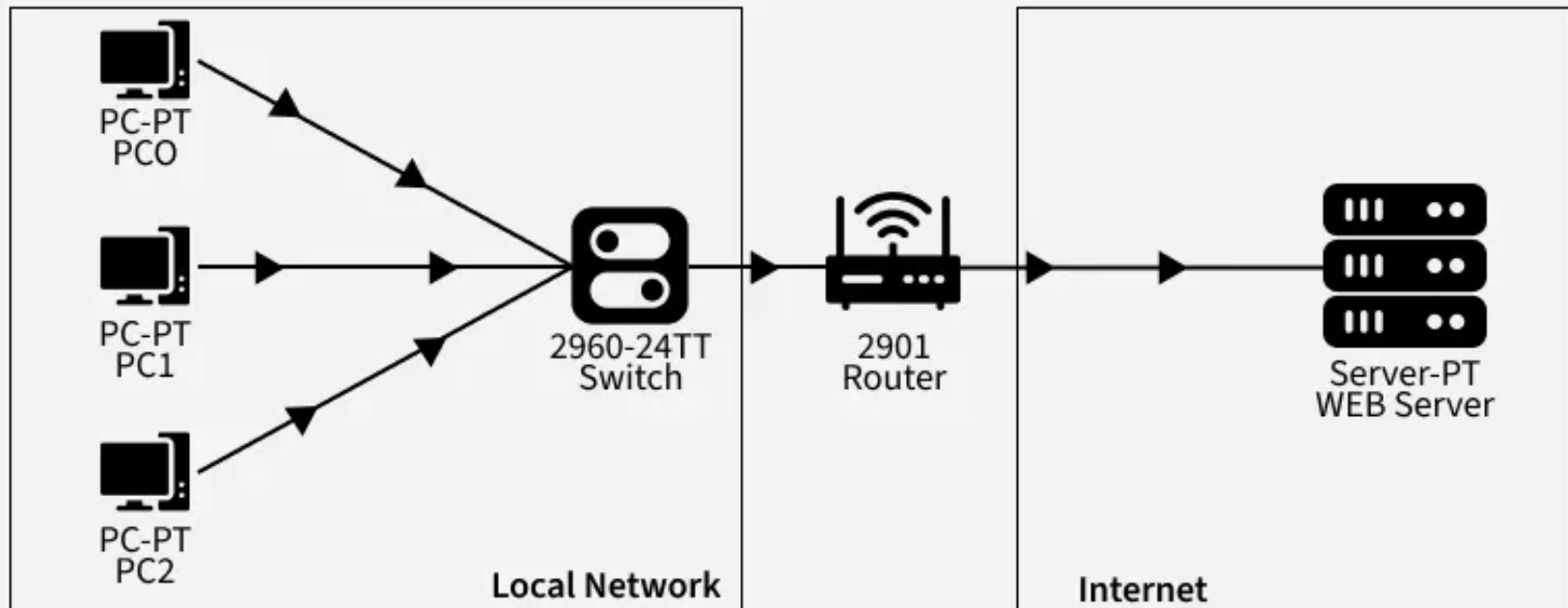


Working

NAT ways to function is listed as below:

- A device **sends a request**, reaches the **NAT enabled router**.
- Router **replaces** the private IP with its public IP and **assigns a unique port**.
- NAT **stores this mapping** in the NAT table.
- When the server responds, NAT uses the stored entry to send the packet to the correct internal device.
- **If NAT runs out of addresses**, i.e., no address is left in the pool configured then the **packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination is sent**.

Working of NAT



Port Number Masking in NAT

– Port Address Translation (PAT)

- Port number masking is required when multiple internal devices use the same port number to communicate with the same external destination.
- Modifies both the private IP address and the source port number
- Creates a unique mapping entry in the NAT table
- Distinguishes traffic from different internal devices
- Ensures that returning packets are delivered to the correct device

Types of NAT

1. Static NAT -

Single private IP address is mapped with a single Public IP address. It is used in Web hosting.

2. Dynamic NAT -

Multiple private IP addresses are mapped to a pool of public IP addresses. It is used when we know the number of fixed users who want to access the Internet at a given point in time.

3. Port Address Translation (PAT) -

This is also known as NAT overload. Many private IP addresses can be translated to a single public IP address. Port numbers are used to distinguish the traffic. This is most frequently used as it is cost-effective as thousands of users can be connected to the Internet by using only one real global (public) IP address.

Thank You !